

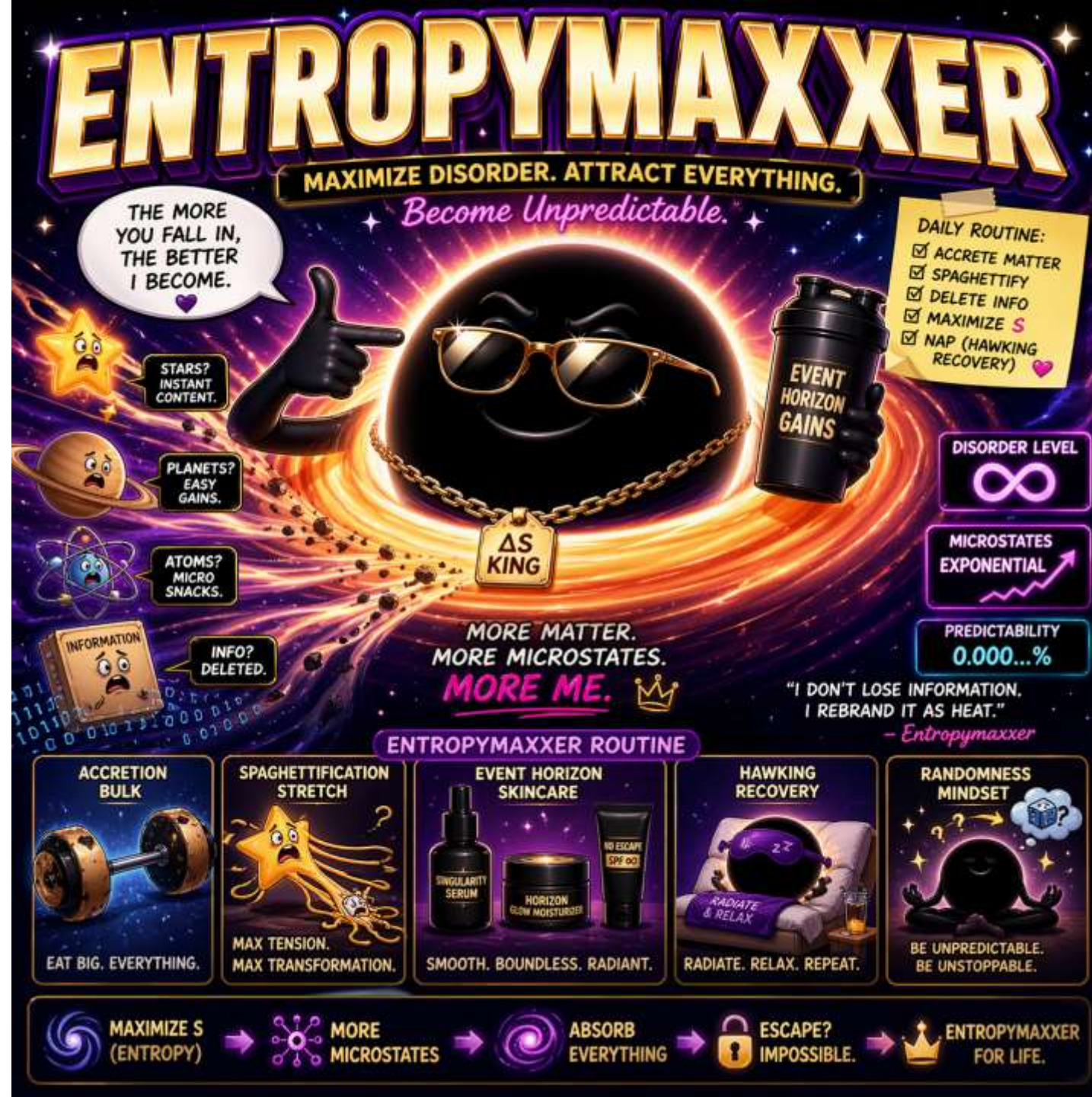
Searching for the Atoms of Spacetime

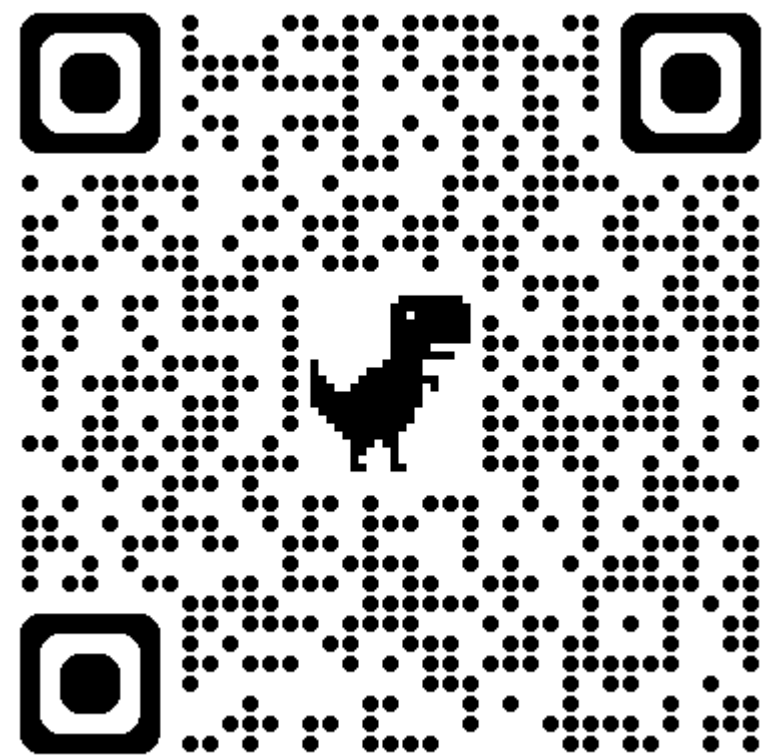
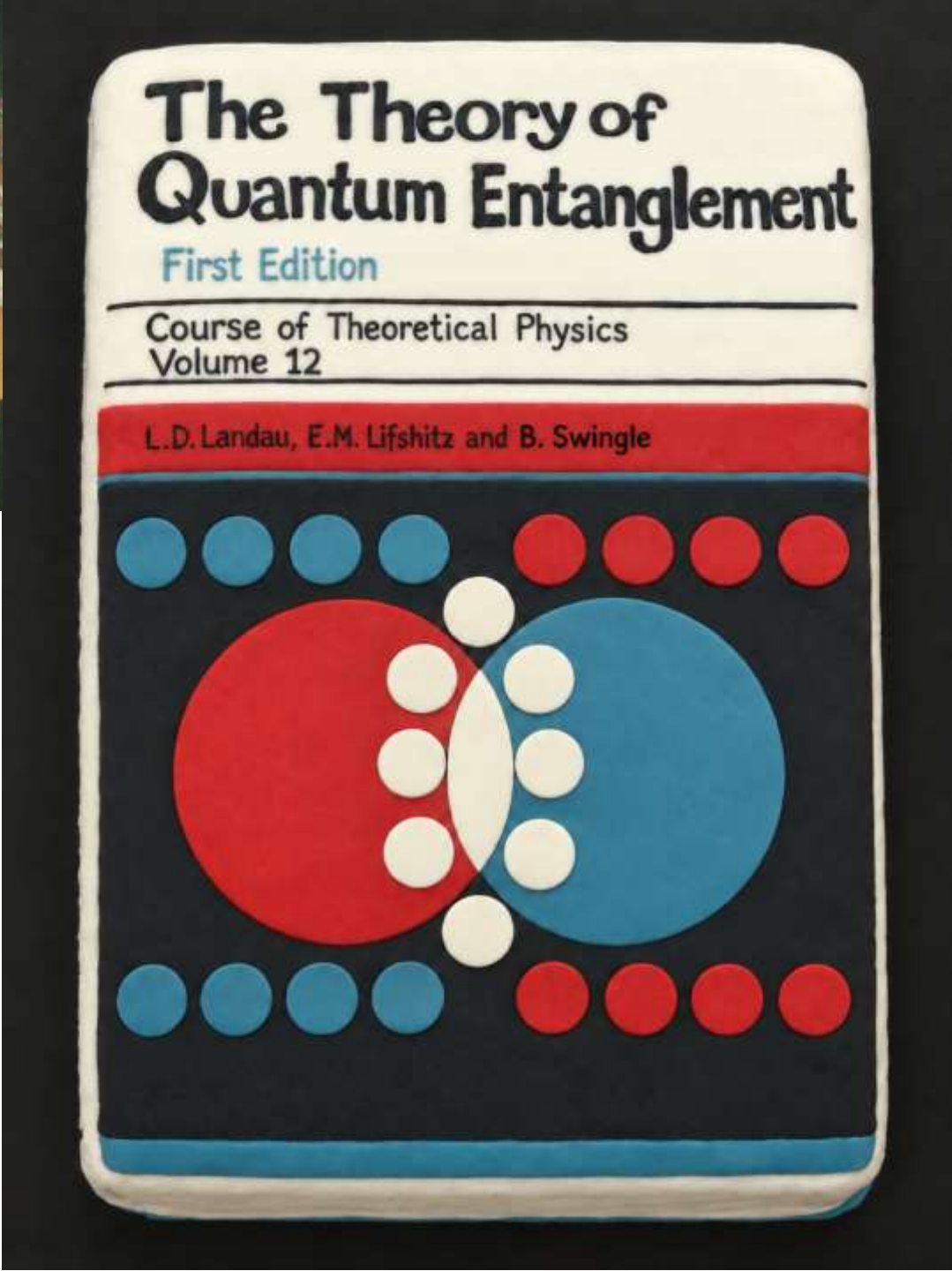
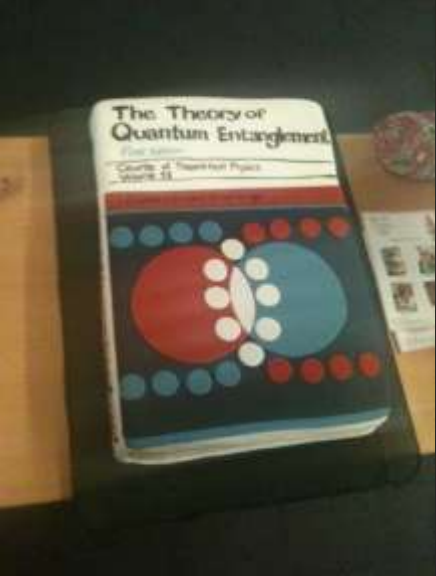
Brian Swingle (Brandeis)

Brandeis Physics Colloquium

Sep 22, 2026

Support from:





physicsmonkey.github.io



PhD students:

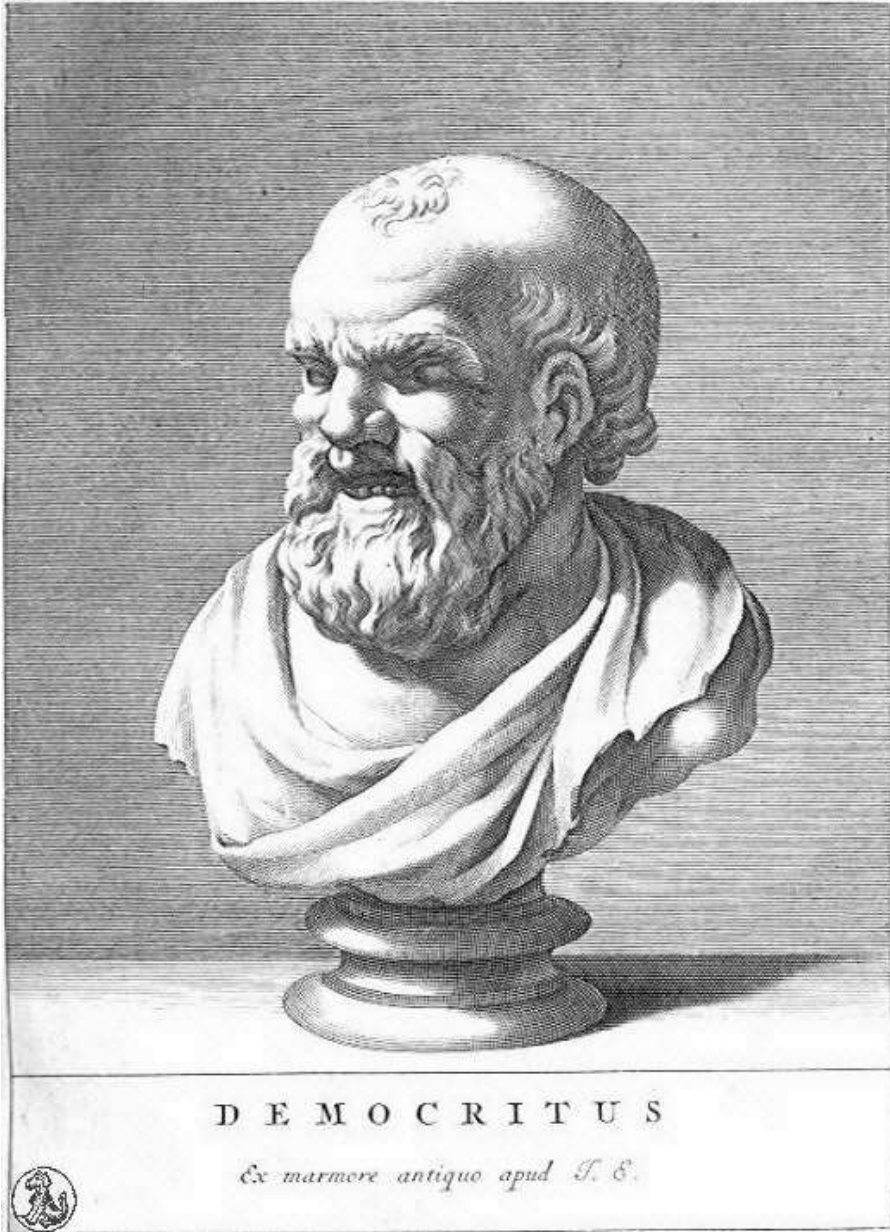
- Yixu Wang → Asst Prof, SIMS
- Gong Cheng → Postdoc, V Tech
- Subhayan Sahu → Postdoc, Perimeter
- Stefano Antonini → Postdoc, Berkeley
- Yongchan Yoo → AI/ML
- Mike Winer → Postdoc, IAS → ARC
- [Valerie Bettaque](#) → Postdoc, Calgary
- [Nadie Li](#)
- [Shiyong Guo](#)
- [Allison Powell](#)

Postdocs:

- Shenglong Xu → Asst Prof, Texas A&M
- Christopher White → Res Sci, NRL
- Chris Baldwin → Asst Prof, U Mich Lansing
- Charles Cao → Asst Prof, Virginia Tech
- Phuc Nguyen → AI/ML
- [Brianna Grado-White](#) → AI/ML
- [Shaokai Jian](#) → Asst Prof, Tulane
- [Greg Bentsen](#) → Asst Prof, William and Mary
- [Martin Sasieta](#) → Postdoc, Berkeley
- [Brian Khor](#)

+ a big thanks to
the extended group

The problem of quantum gravity



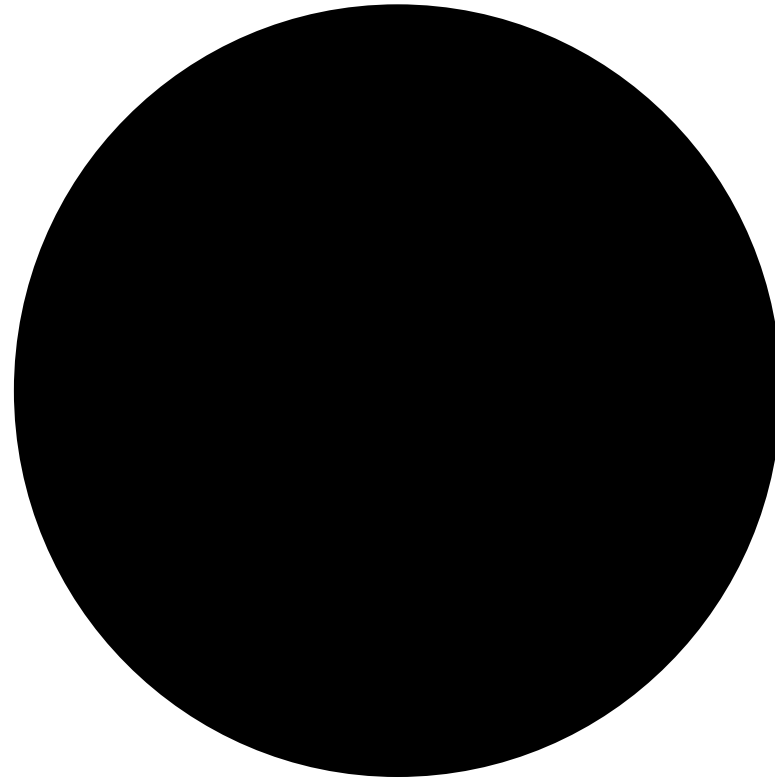
“By convention sweet and by convention bitter, by convention hot, by convention cold, by convention color; but in reality atoms and void.”



“If, in some cataclysm, all of scientific knowledge were to be destroyed, ..., what statement would contain the most information in the fewest words? I believe it is the atomic hypothesis ... that all things are made of atoms—little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another.”

Why is the void interesting?

Our best theory of spacetime and gravity, general relativity, predicts the existence of black holes (BH) ...

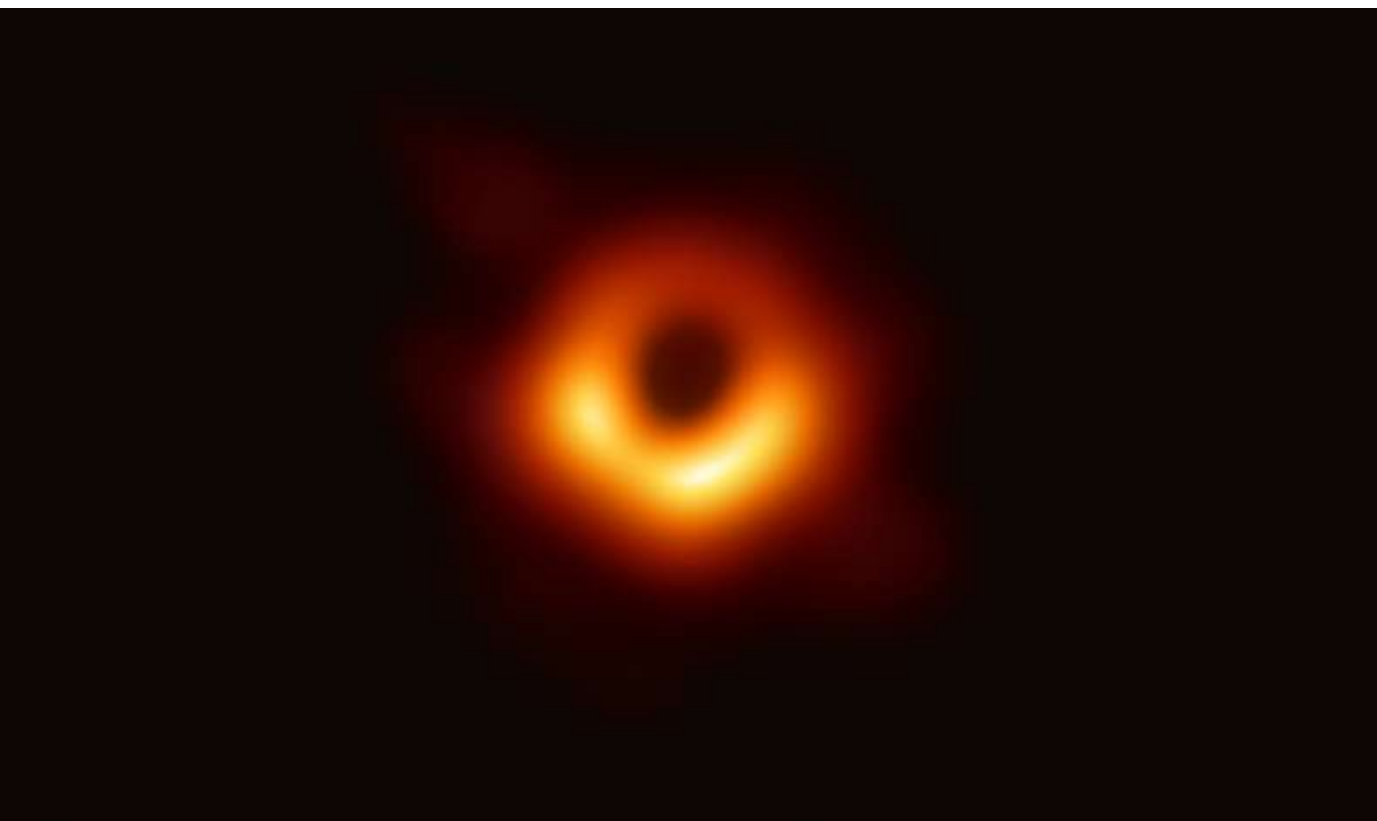


$$R = 2G_N M$$

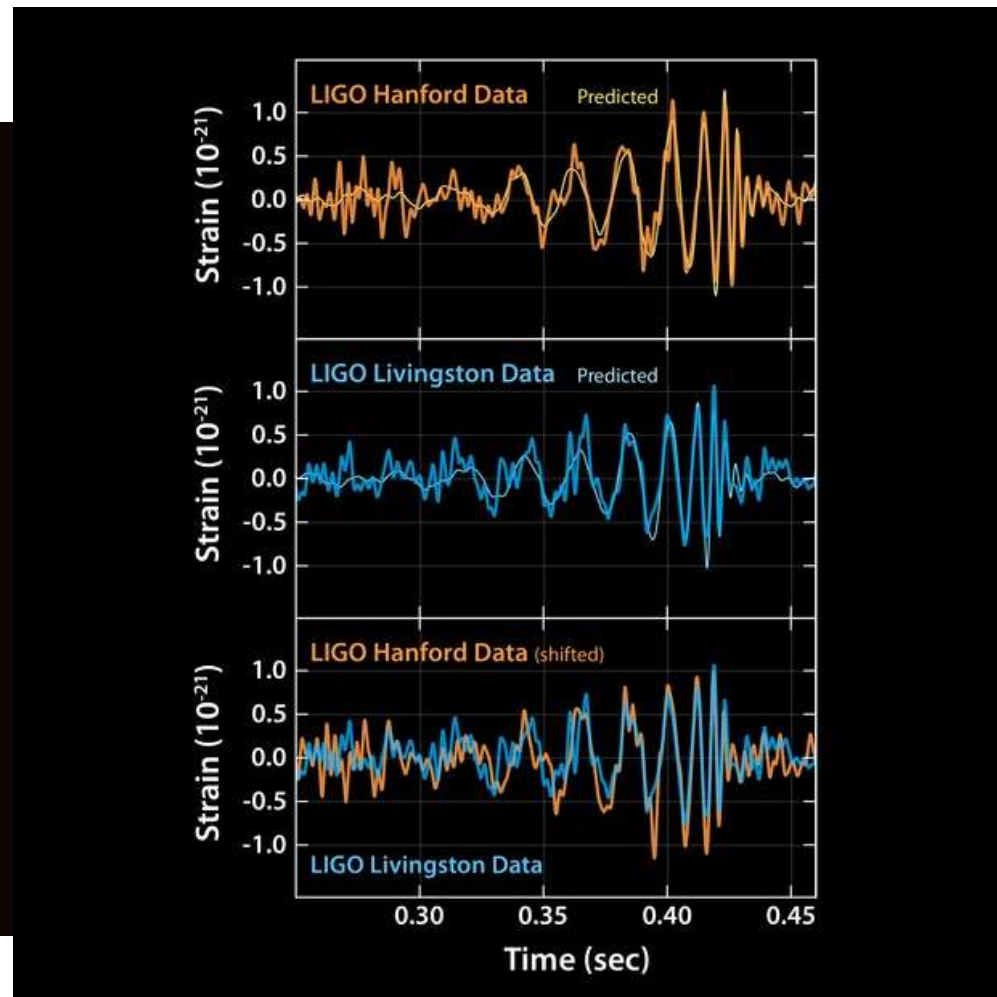
[Schwarzschild 1916]

use units where $c=1$, $k_B=1$
(speed of light, Boltzmann)

... for which we have strong observational evidence



[Supermassive black hole in M87 – EHT, 2019]

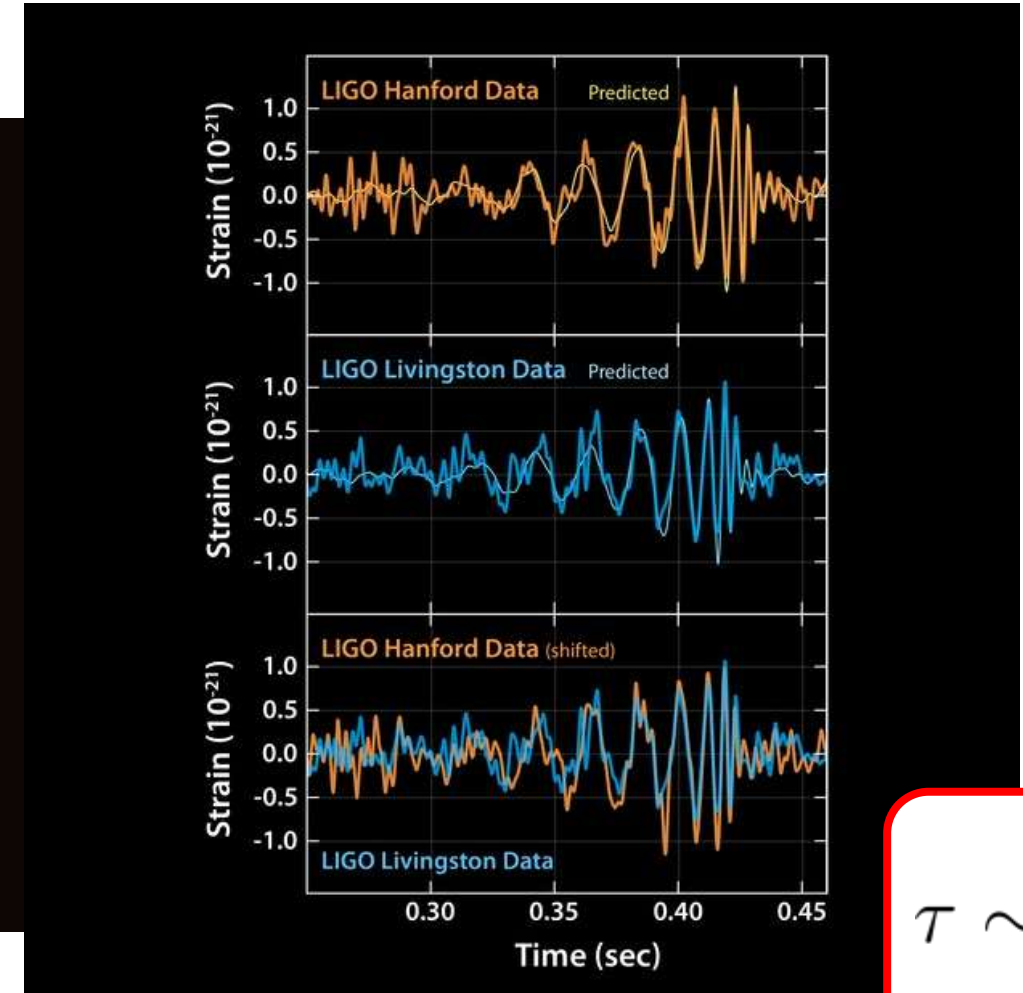


And black holes can seemingly be understood as strongly interacting quantum many-body systems!!!

$$S = \frac{\text{Area}}{4G_N\hbar} \quad [\text{Bekenstein, Hawking}]$$

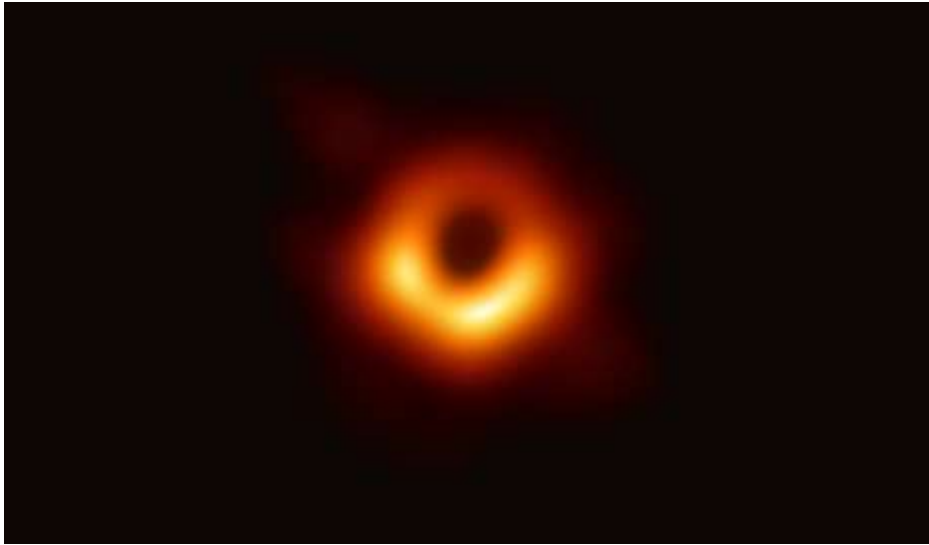


[Supermassive black hole in M87 – EHT, 2019]

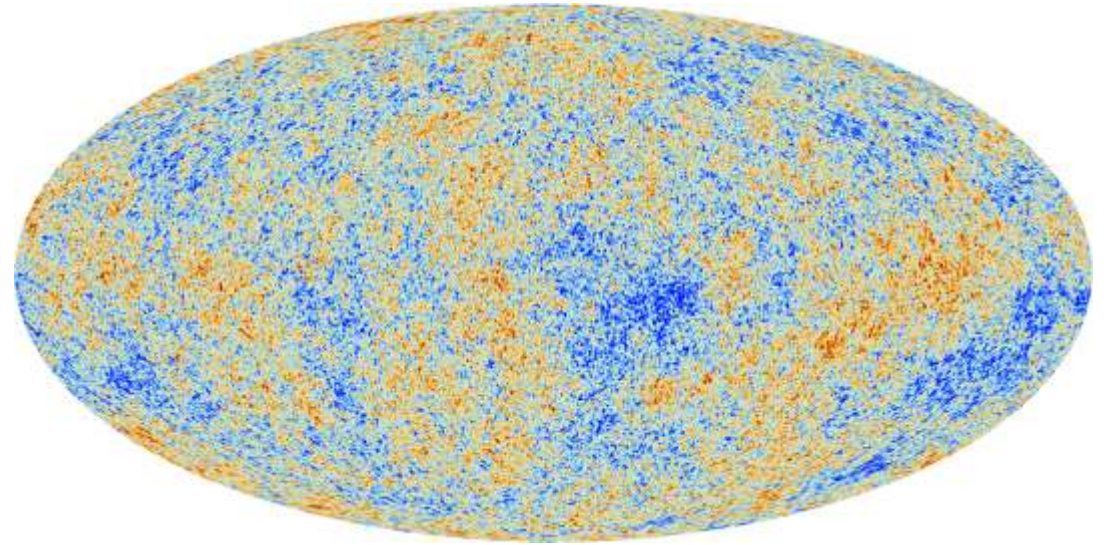


$$\tau \sim \frac{\hbar}{T}$$

In fact, black holes are predicted to be the most entropic objects in the universe ...



Entropy M87* $\sim 10^{96}$ bits



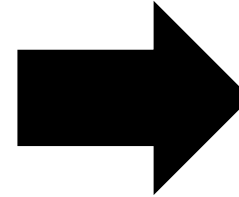
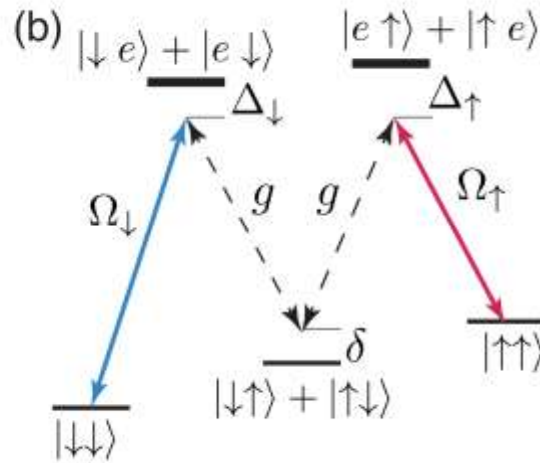
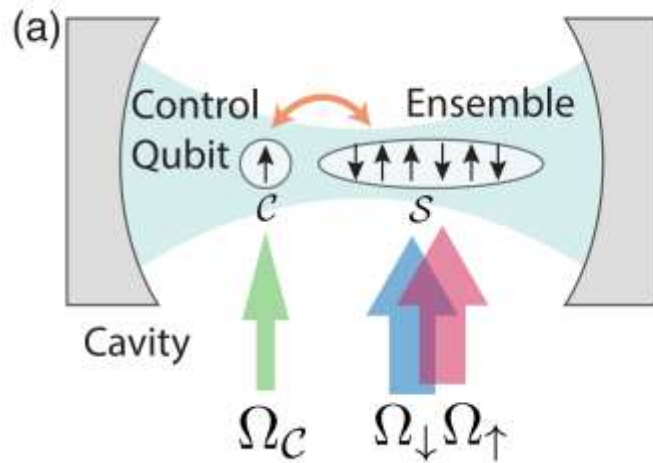
Entropy of CMB photons $\sim 10^{89}$ bits

... except perhaps for the cosmic horizon
which would translate to $\sim 10^{122}$ bits!

Why study quantum gravity

- We just want to know how the universe is, plus it's interesting and fun! QG naturally brings together many subjects: quantum field theory, geometry, information theory, statistical physics, ...
- We will one day have direct experimental access to QG effects, and the in meantime we have well-formed puzzles to guide us
- It has many connections, e.g. to math and philosophy
- Remarkably, we can potentially study QG phenomena in table-top experiments with programmable quantum systems

Why study quantum gravity



- Remarkably, we can potentially study QG phenomena in table-top experiments with programmable quantum systems

What are the atoms of spacetime?

Today: **four broad efforts in my group to address this question:**

- How can we build empty space and wormholes?
- How can we build black holes?
- How can we probe a black hole's dynamics?
- How can we build cosmologies?

We have found that the microstructure of spacetime involves a whole host of structures including entanglement, randomness, computational complexity, error correction, and so much more ...

A toy model: holography

Clues: holographic nature of gravity

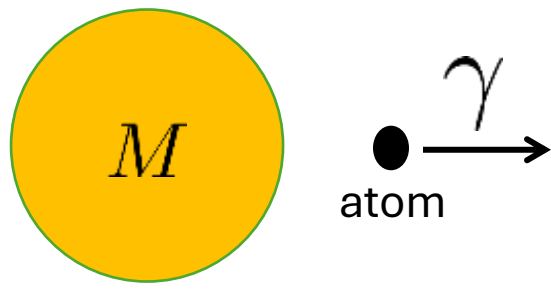
1. The Hamiltonian is measurable at infinity

$$F = \frac{G m_{\text{test}}}{r^2} \left(\frac{E}{c^2} \right) \quad [\text{Newton}]$$

2. Black holes have entropy proportional to their area

$$S_{BH} = \frac{c^3 A_{BH}}{4\hbar G_N} \quad [\text{Bekenstein-Hawking}]$$

3. Gravitational redshift

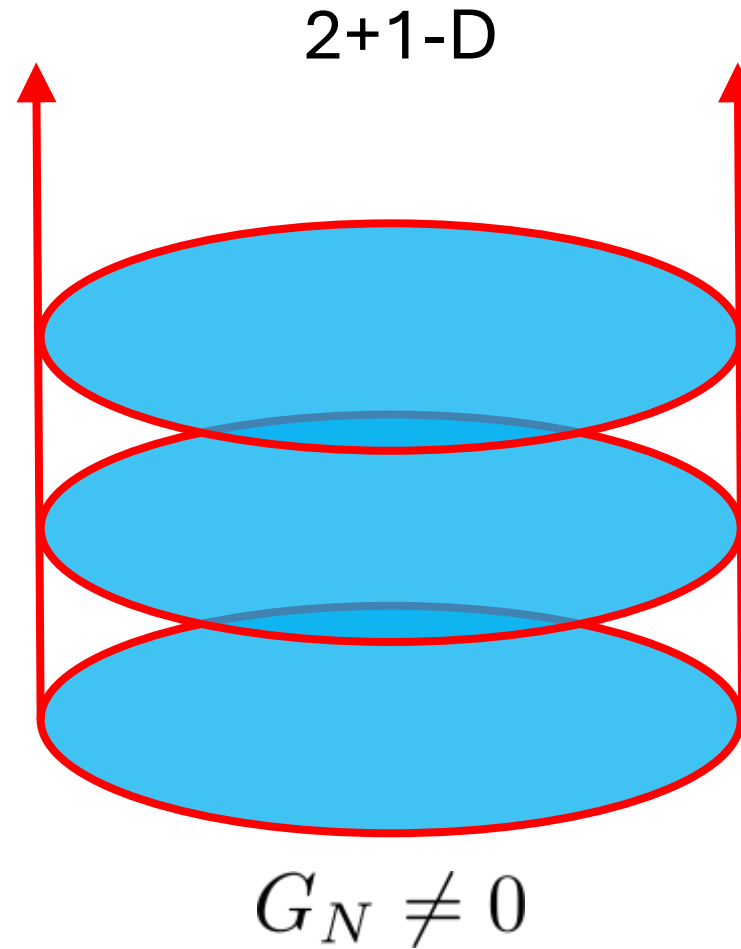
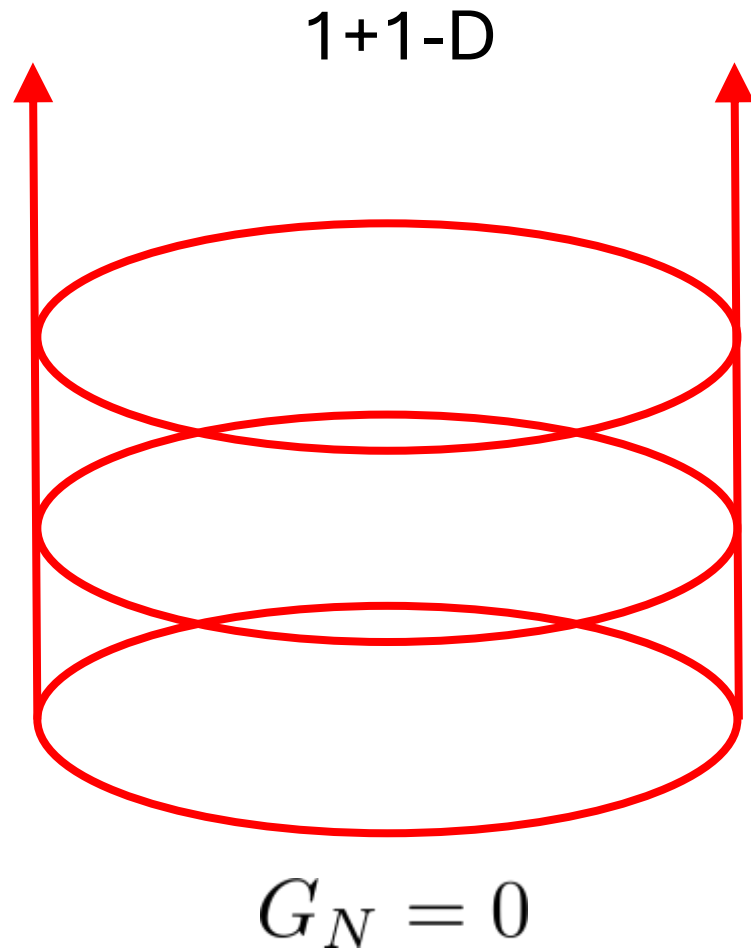

$$E_{\infty} = E_0 \sqrt{1 - \frac{2GM}{rc^2}} \quad [\text{Einstein}]$$

To study quantum gravity, we will sacrifice astrophysical realism for a simpler **holographic** setting which retains (1) the unique features of black hole entropy and (2) the fast dynamics

One great advantage of this setting is that it connects to table-top physics



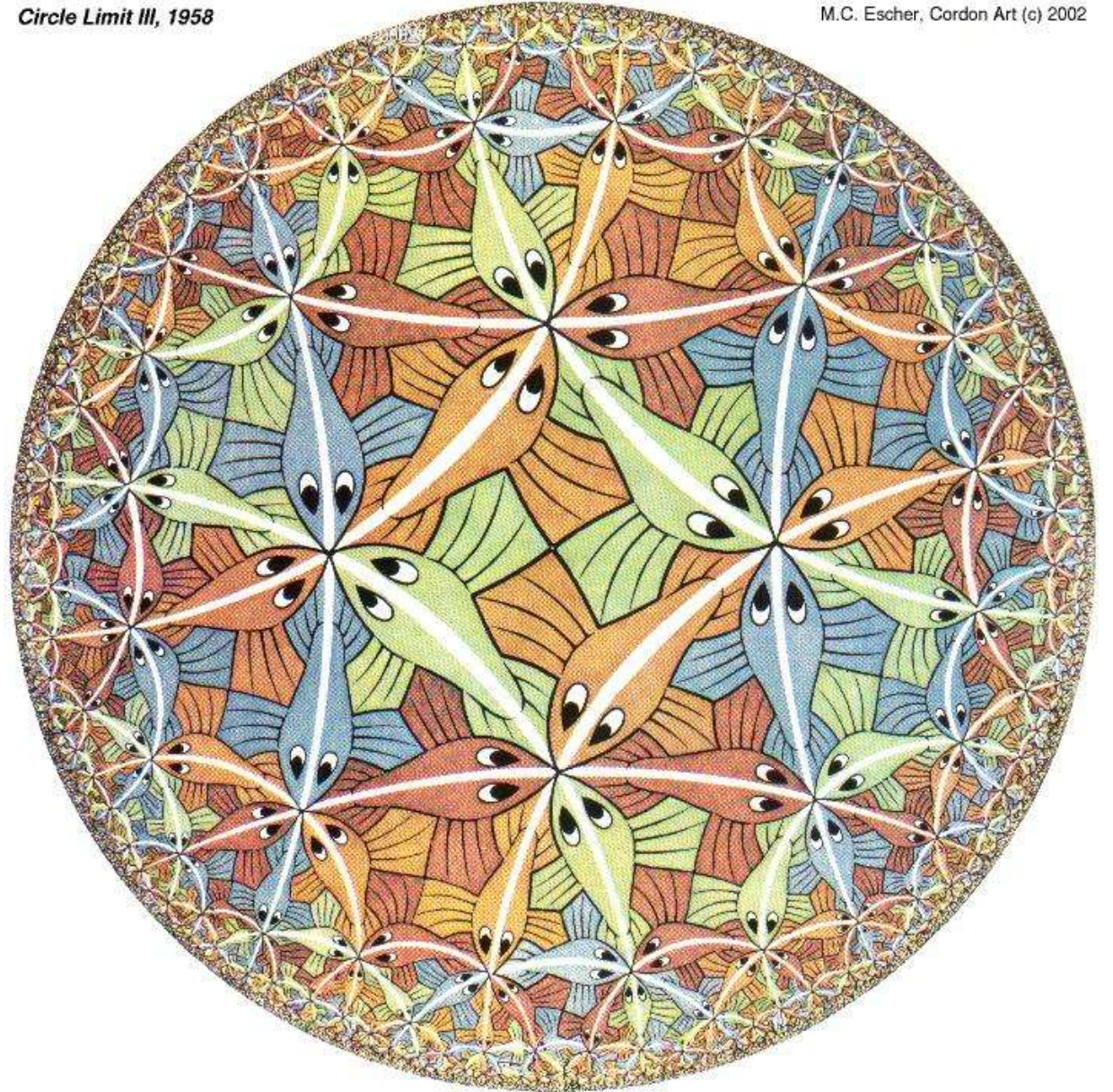
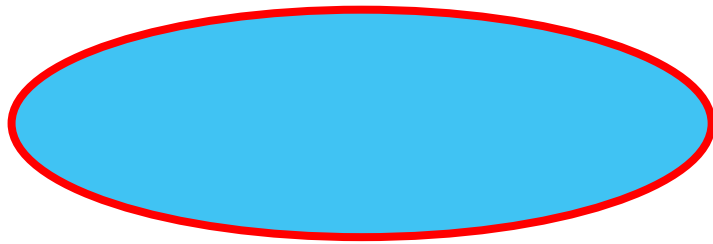
Anti de Sitter space / conformal field theory (AdS/CFT or holographic duality)



[Maldacena]

Circle Limit III, 1958

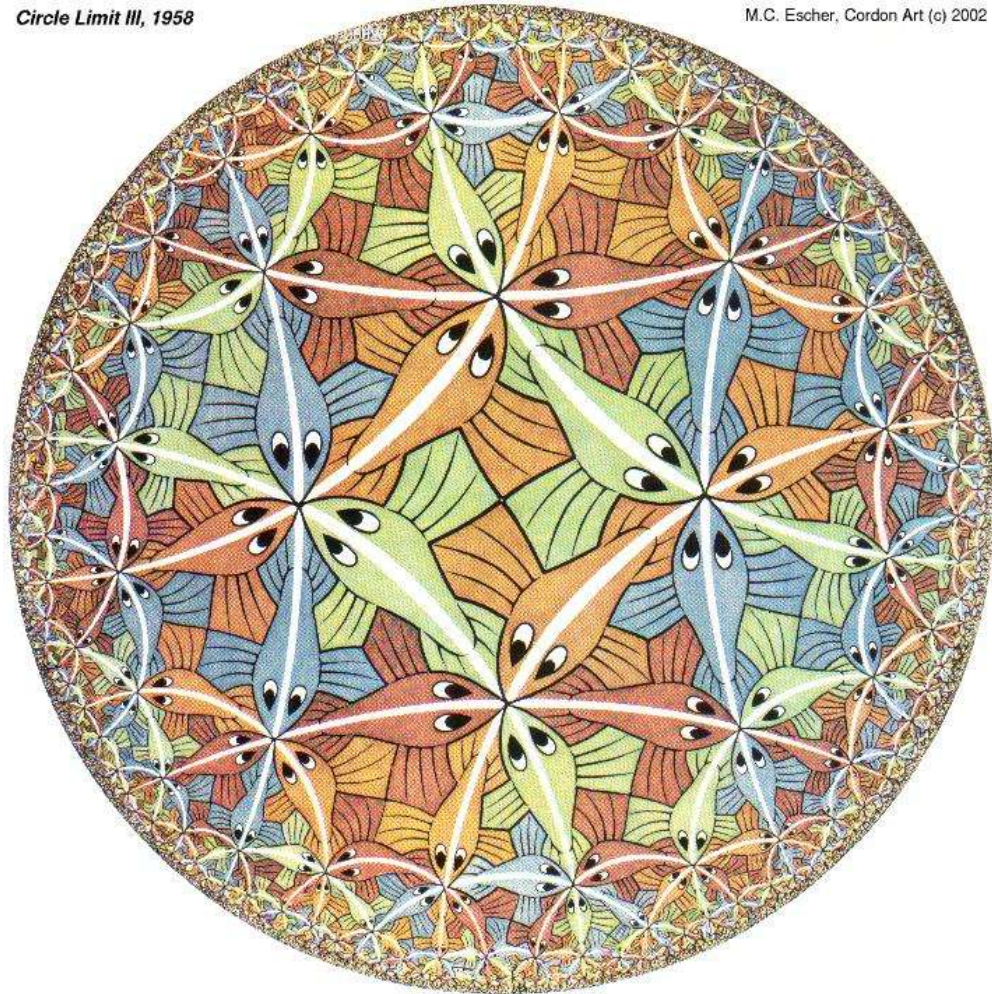
M.C. Escher, Cordon Art (c) 2002



AdS and an AdS black hole

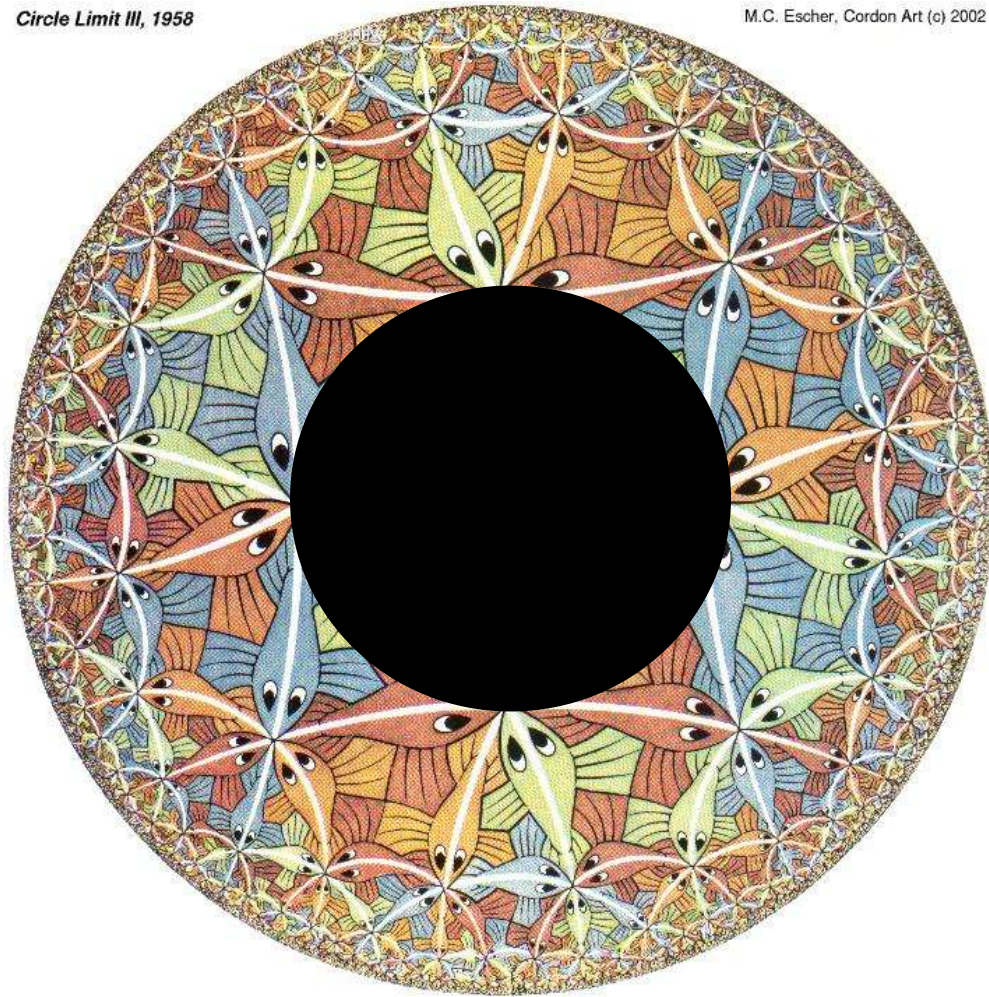
Circle Limit III, 1958

M.C. Escher, Cordon Art (c) 2002



Circle Limit III, 1958

M.C. Escher, Cordon Art (c) 2002



Entanglement

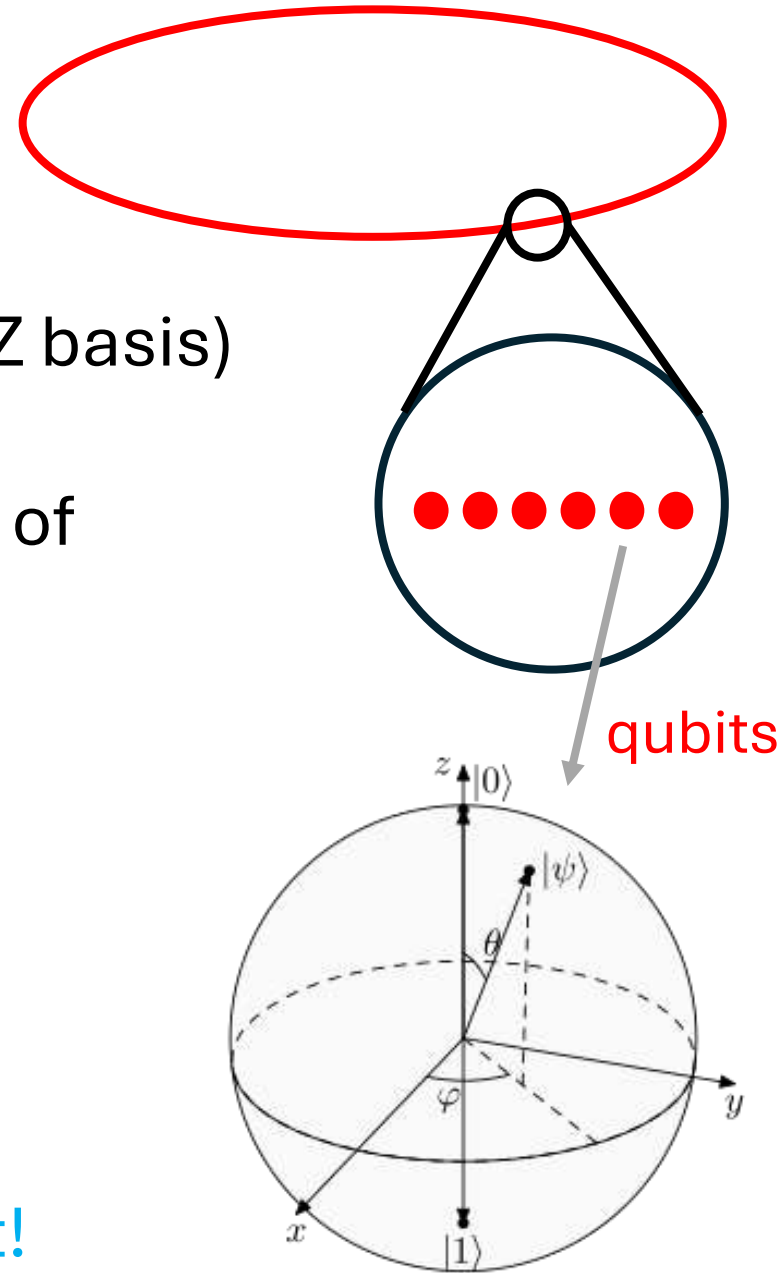
Four states: 00 and 01 and 10 and 11 (Z basis)

Superposition: any linear combination of states is also allowed

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \neq |\phi_1\rangle |\phi_2\rangle$$

EPR or Bell pair

Definition of entanglement!



Ryu-Takayanagi entanglement formula

S = Von Neumann entropy

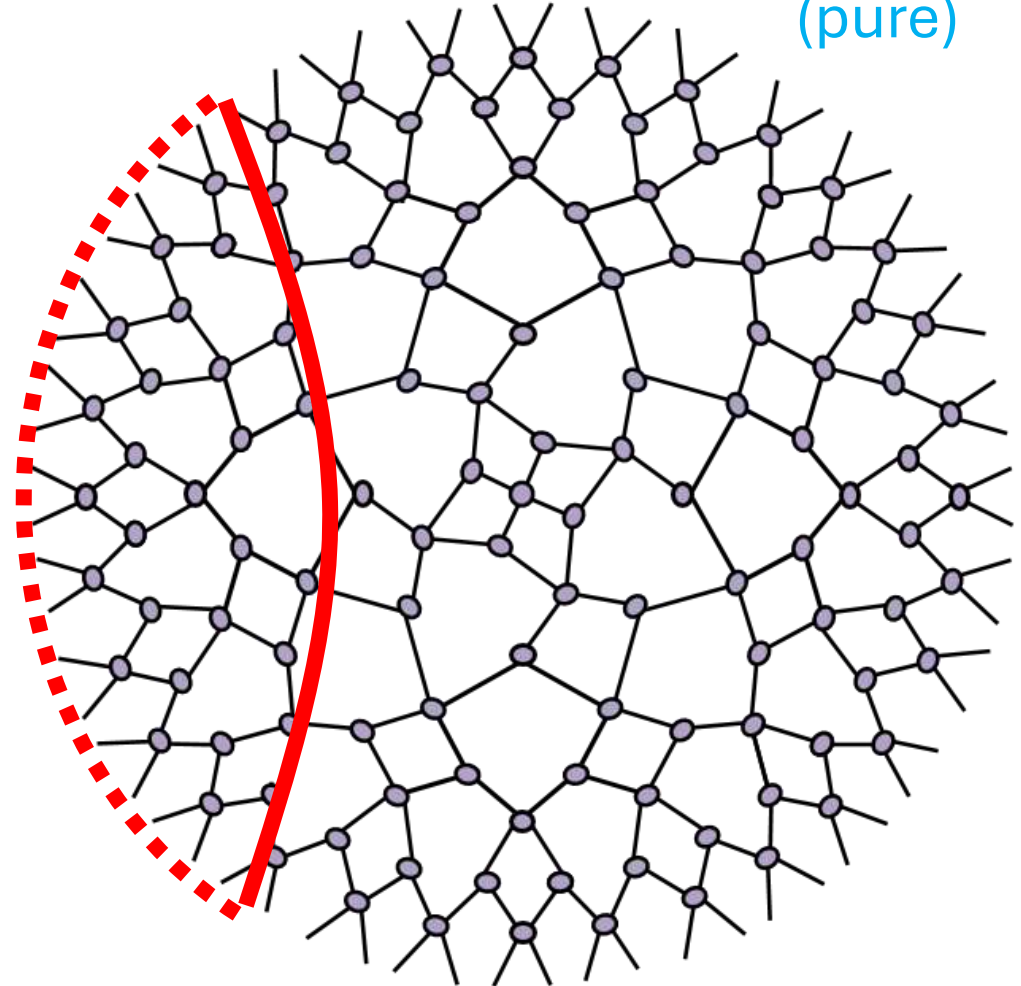
M.C. Escher, Goudon Art (c) 2002

A

$$S(A) = \frac{\text{Minimal Area}}{4\hbar G_N}$$

[Ryu-Takayanagi]

ground state
(pure)



[S, Pastawski et al., Hayden et al., ...]

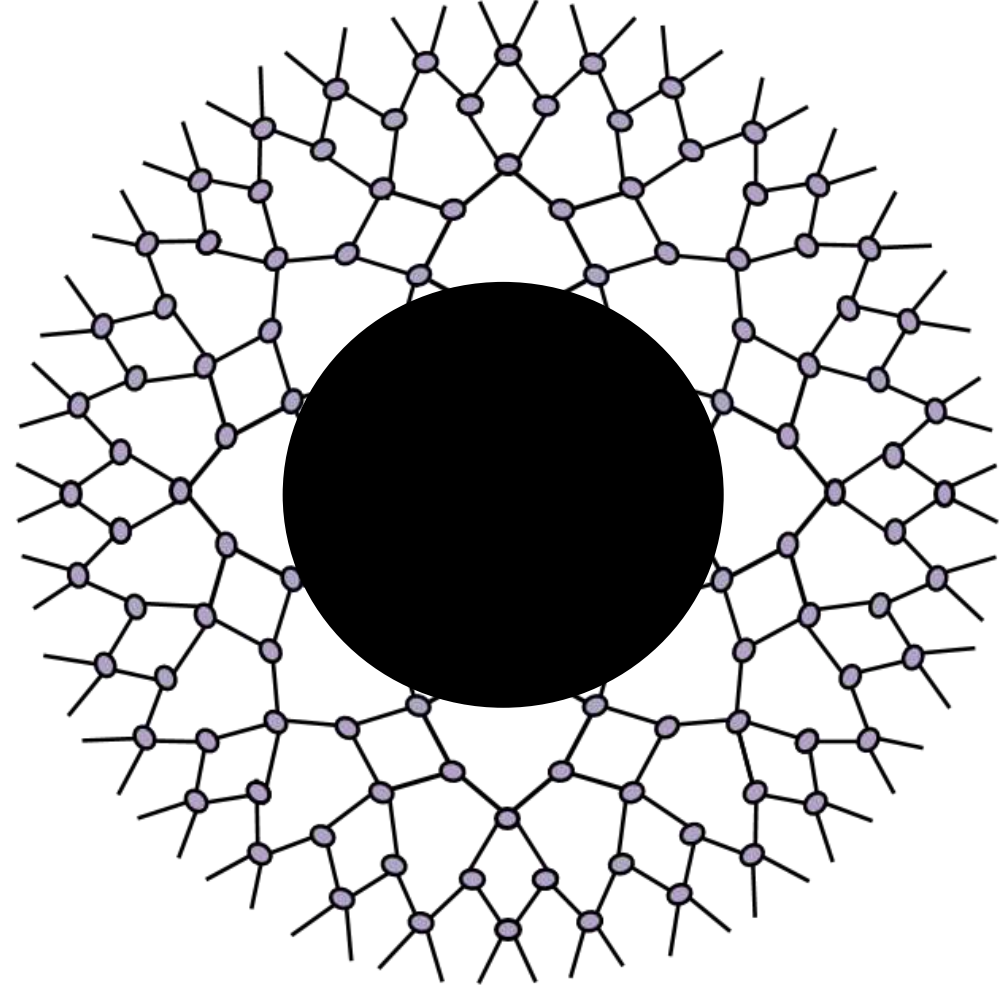
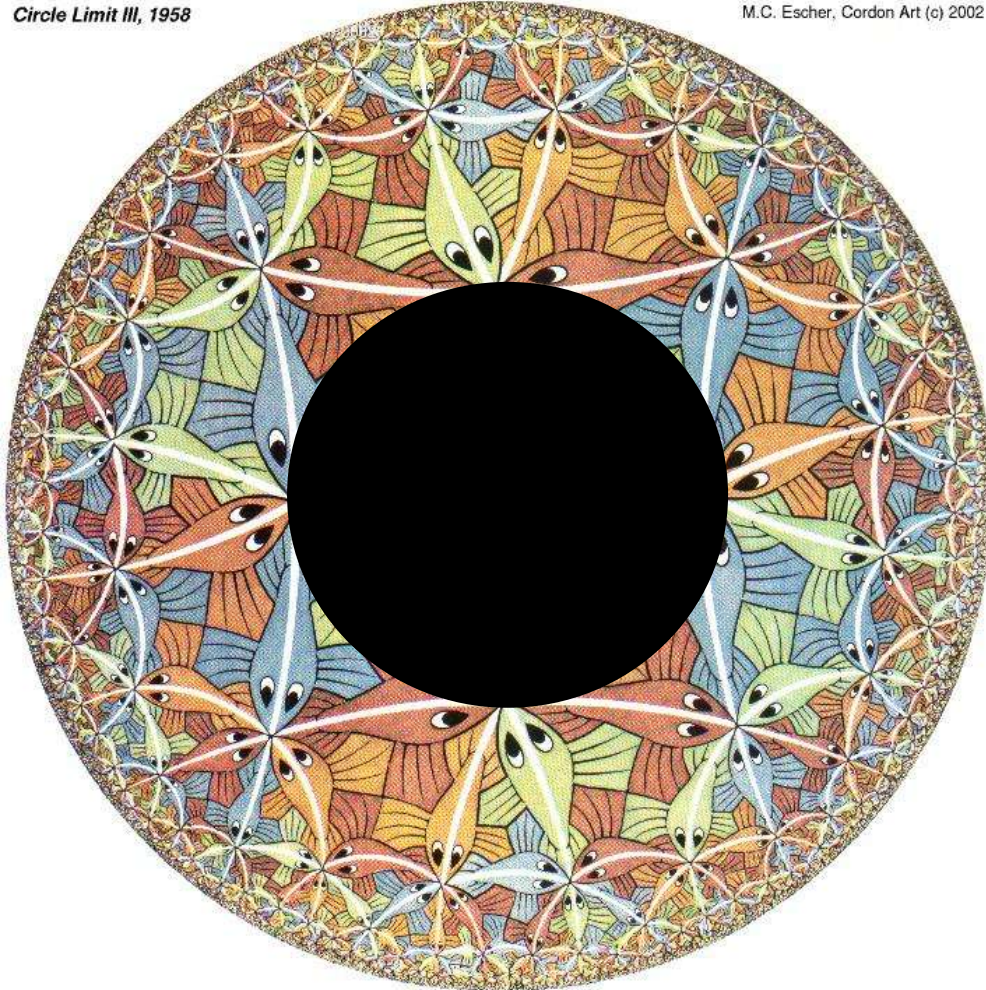
Building a black hole

$$\rho = \frac{e^{-\beta H}}{Z(\beta)}$$

thermal Gibbs
state (mixed)

Circle Limit III, 1958

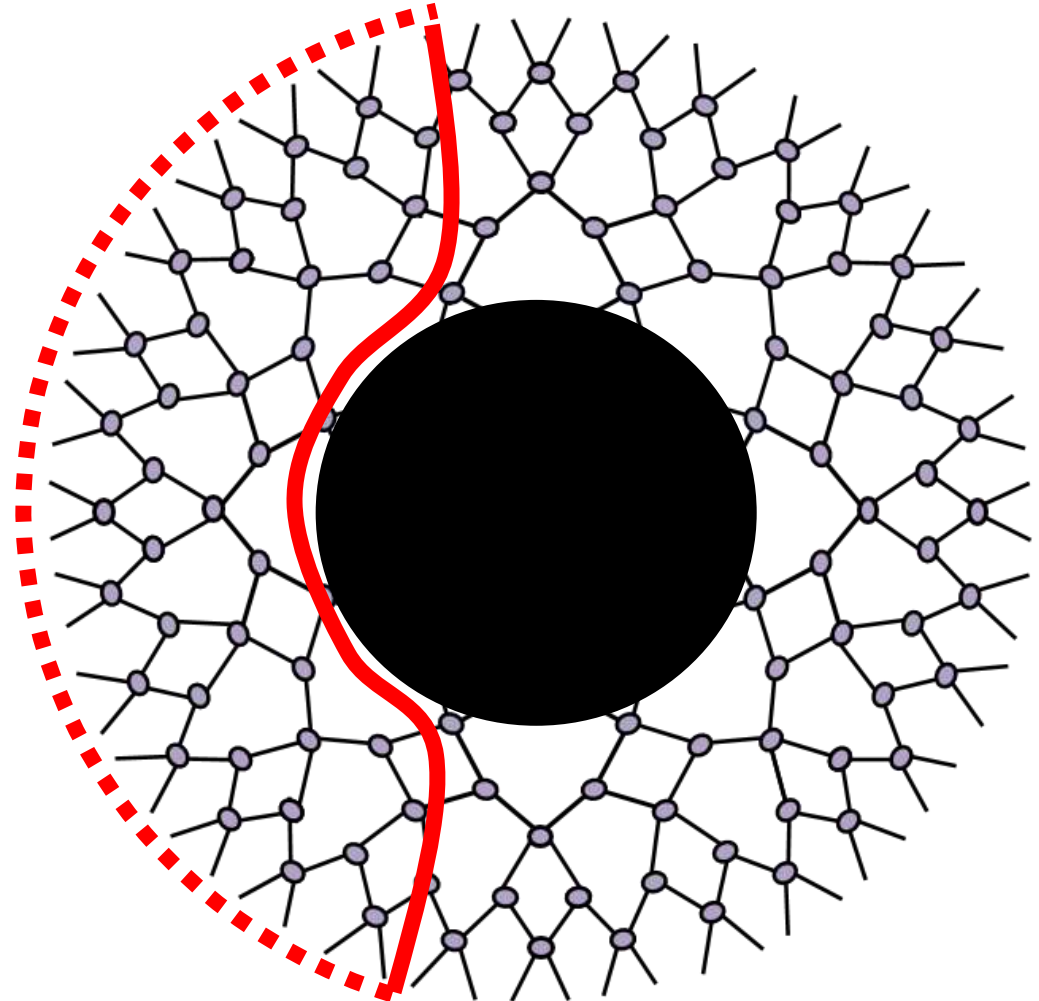
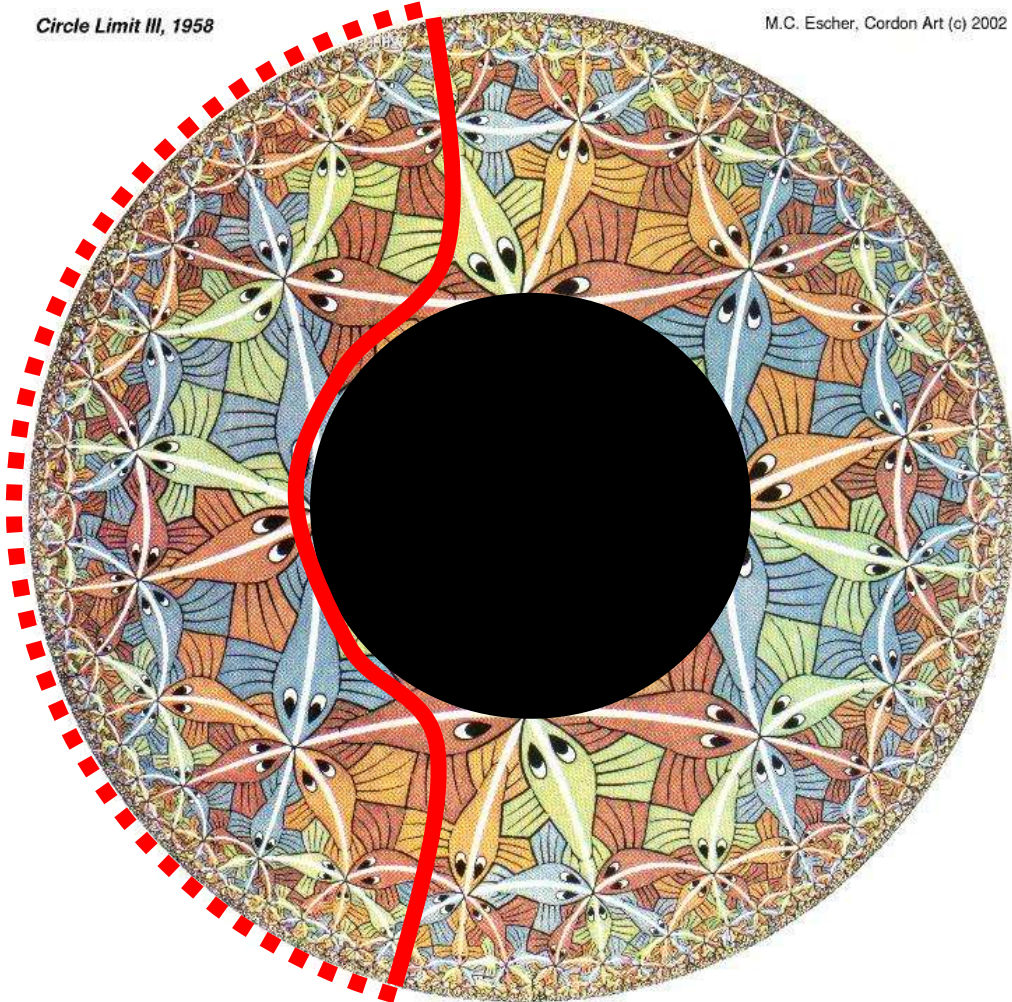
M.C. Escher, Cordon Art (c) 2002



Black hole entropy

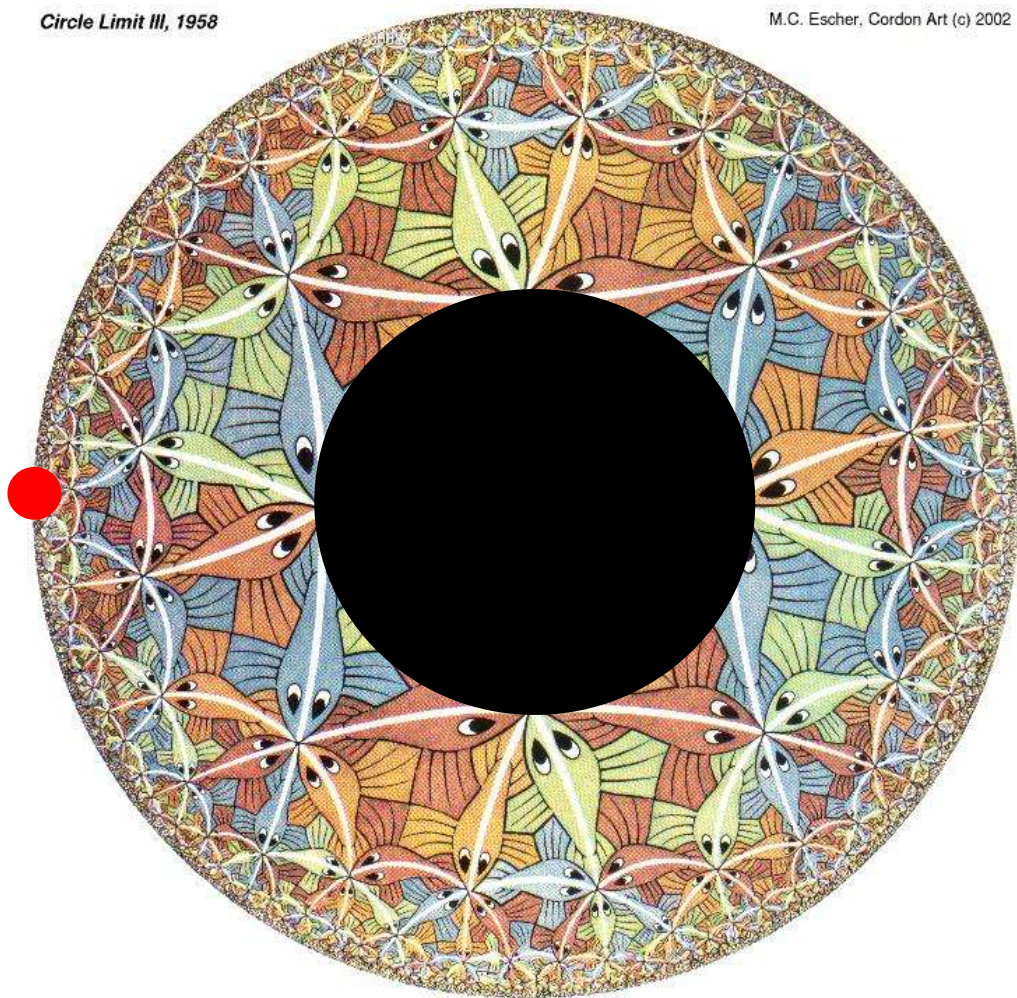
Circle Limit III, 1958

M.C. Escher, Gorden Art (c) 2002



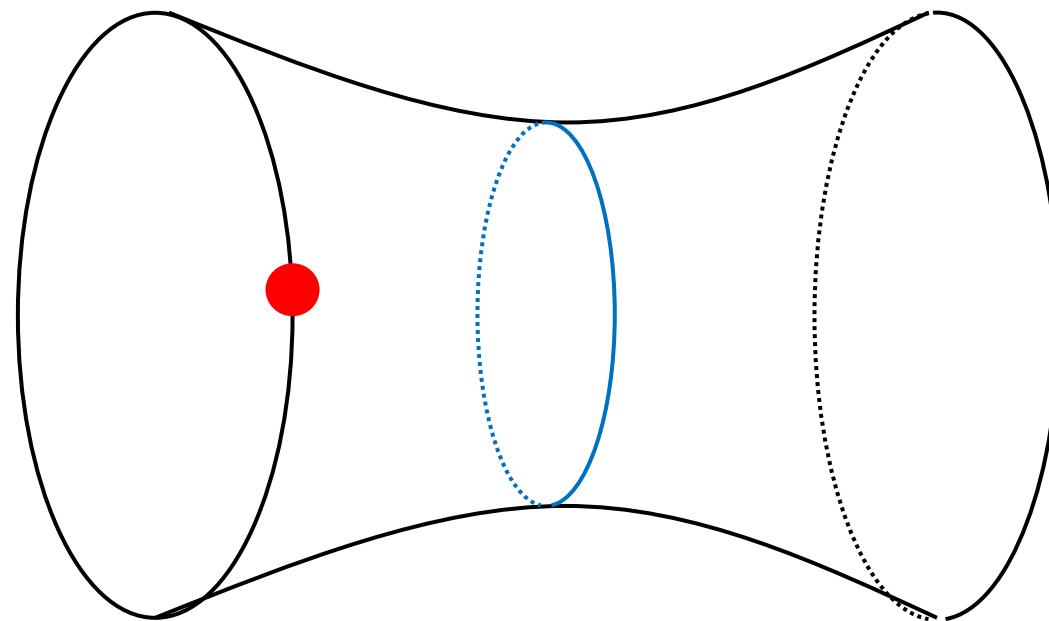
Circle Limit III, 1958

M.C. Escher, Cordon Art (c) 2002



~ mixed state

Two-sided wormhole



~ purification

Geometry from entanglement

Building Holographic Entanglement by Measurement



Jonathan Jeffrey



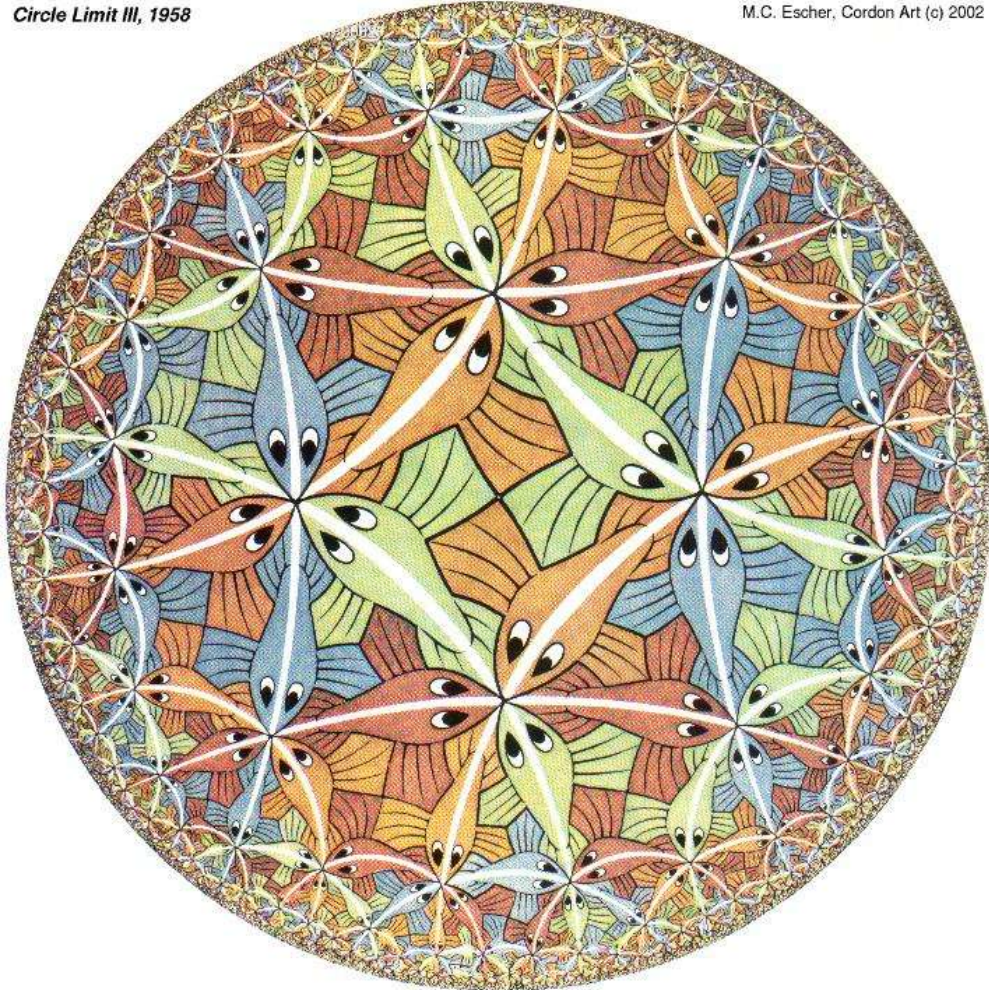
Monika Schleier-Smith



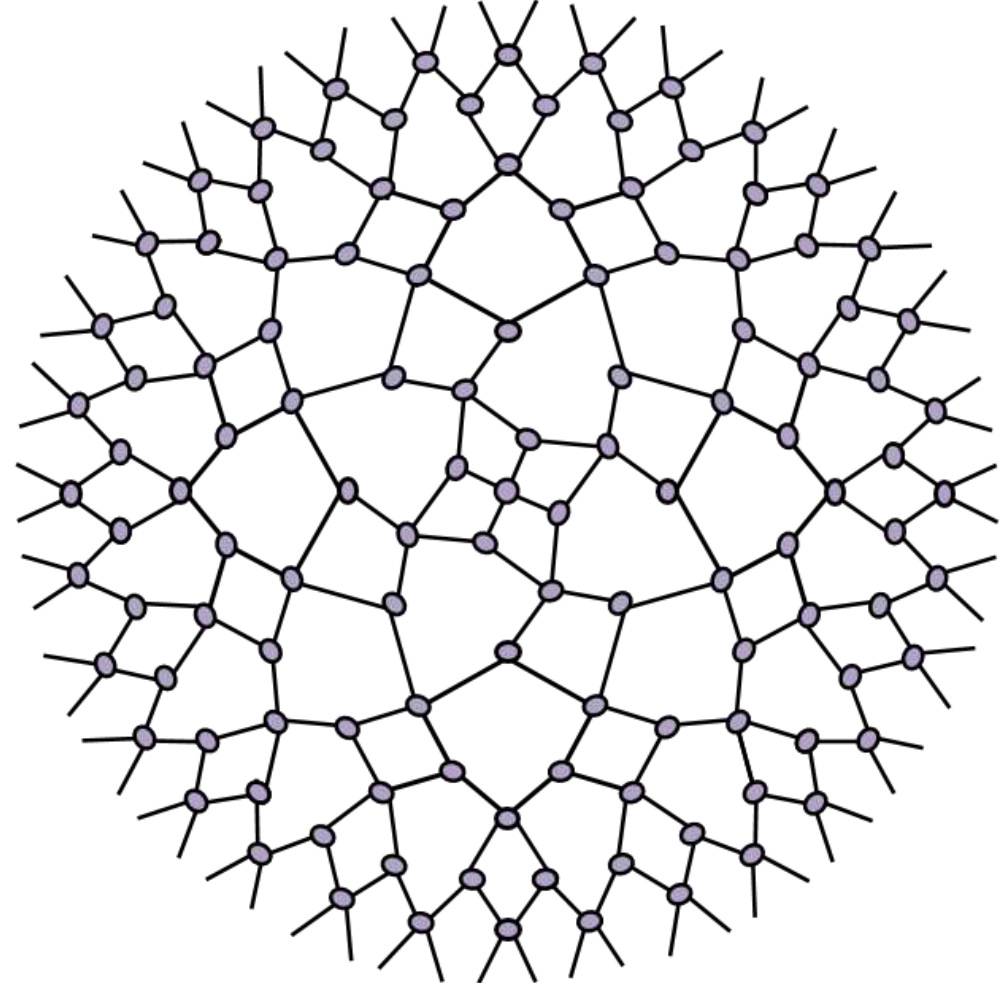
Lucien Gandarias

Building spacetime from entanglement

Circle Limit III, 1958



M.C. Escher, Goudon Art (c) 2002

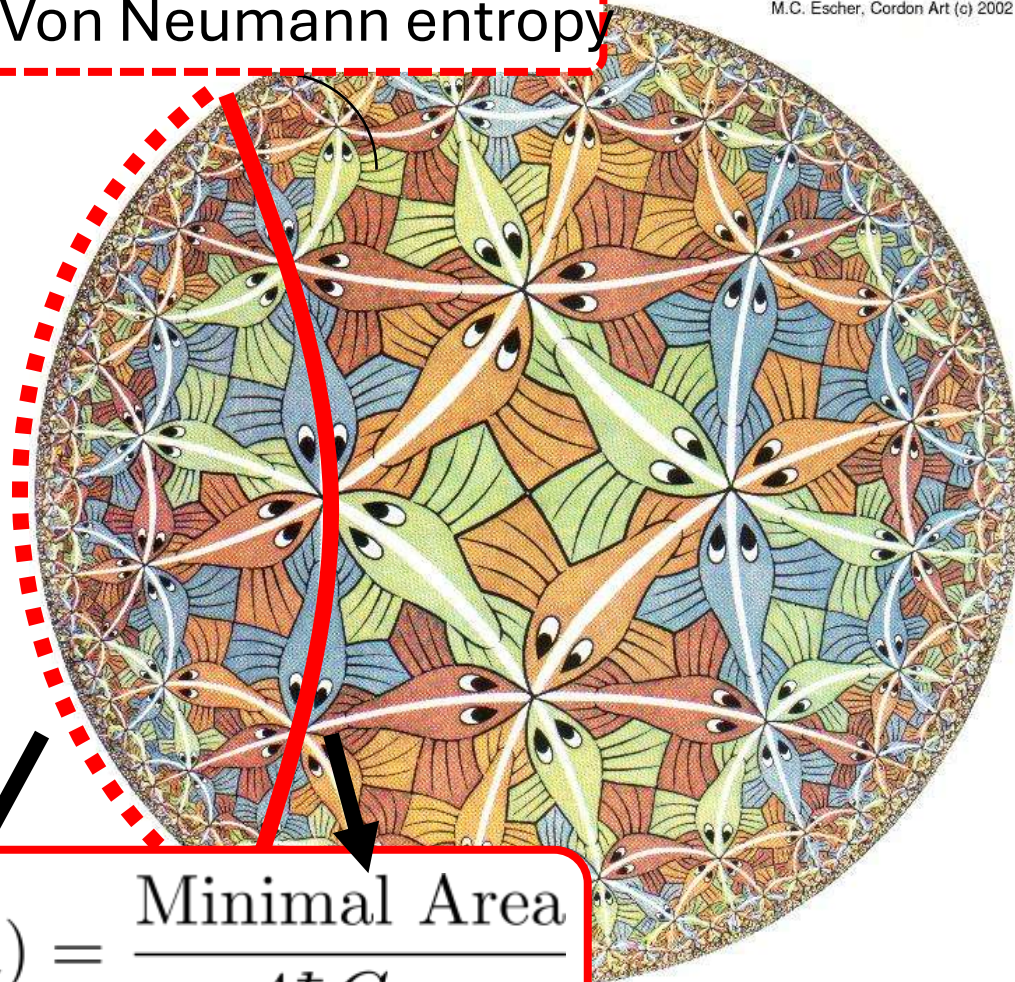


[S, Vidal-Evenbly]

Ryu-Takayanagi entanglement formula

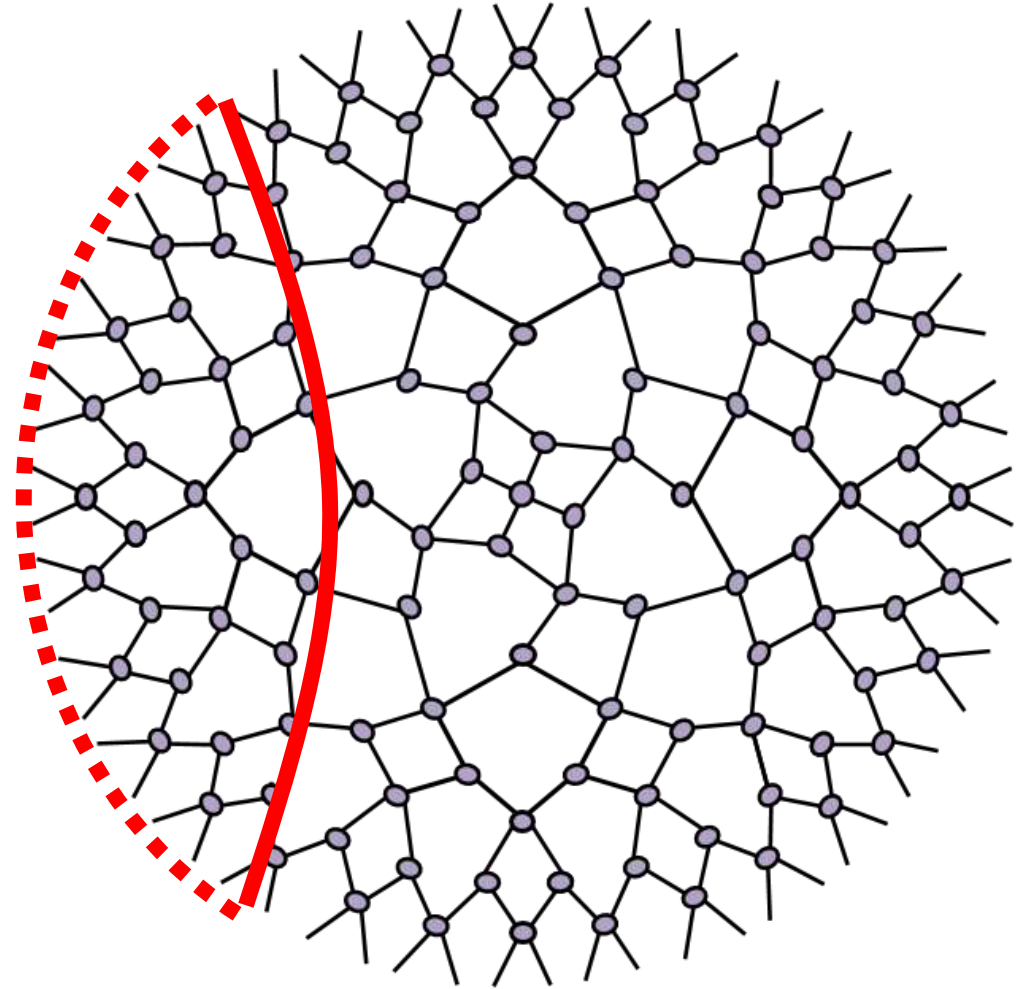
S = Von Neumann entropy

M.C. Escher, Goudon Art (c) 2002



$$S(A) = \frac{\text{Minimal Area}}{4\hbar G_N}$$

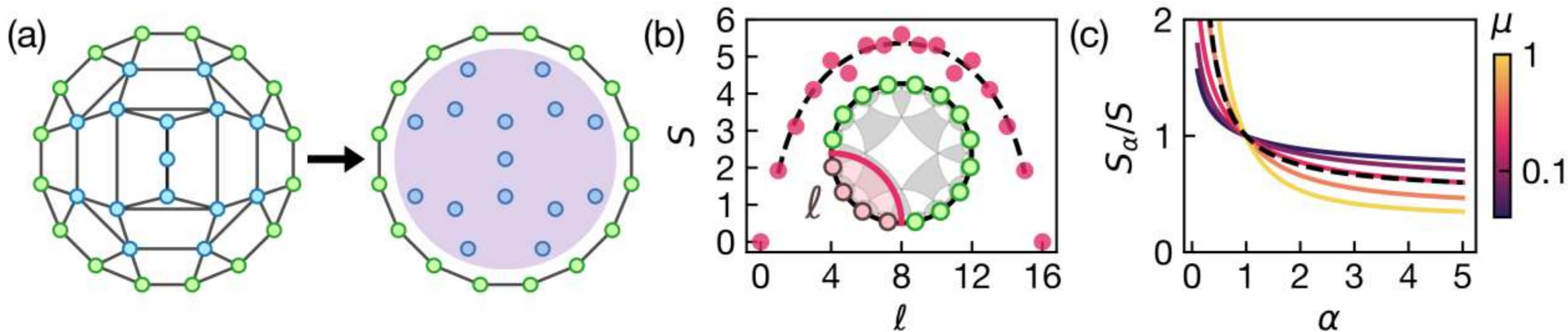
[Ryu-Takayanagi]



[S, Pastawski et al., Hayden et al., ...]

Our proposal and key results

- Inspired by experimentally motivated constraints, we exhibit a simple setting with Ryu-Takayanagi-like (RT) entanglement
- Start with a graph which discretizes a chosen geometry, quench for $O(1)$ time, and measure the bulk $\rightarrow$ a pure boundary state



Key ingredients

- Degrees of freedom are sites that each contain a single canonical pair (x,p) {for example, each site may be a collection of atoms with a collective angular momentum, $[S_x, S_y] = i\hbar^2 s + i\hbar(S_z - s\hbar)$ }
- Sites have arbitrary interaction connectivity {this could be achieved by coupling atomic ensembles with light}
- Any Gaussian operations are allowed: phase space shifts, evolution with quadratic Hamiltonians, and measurements of x,p
- These ingredients are motivated with atomic ensembles in a cavity in mind, but there are other experimental platforms, e.g. photons

Setup

- Let's use the following protocol:
 1. Choose a graph $G=(V,E)$ with bulk and boundary sites comprising V
 2. The initial wavefunction is a product of Gaussians for each site in V
 3. The system is quenched: evolve with $H_q = \frac{1}{2} \sum_{ab} x_a J_{ab} x_b$ for unit time where J is set by the (weighted) adjacency matrix of G
 4. Measure the bulk sites in the momentum basis, and (optionally) apply local phase space displacements conditioned on the outcomes
- We analyze the entanglement entropy, Renyi entropy, and correlations of the resulting pure states

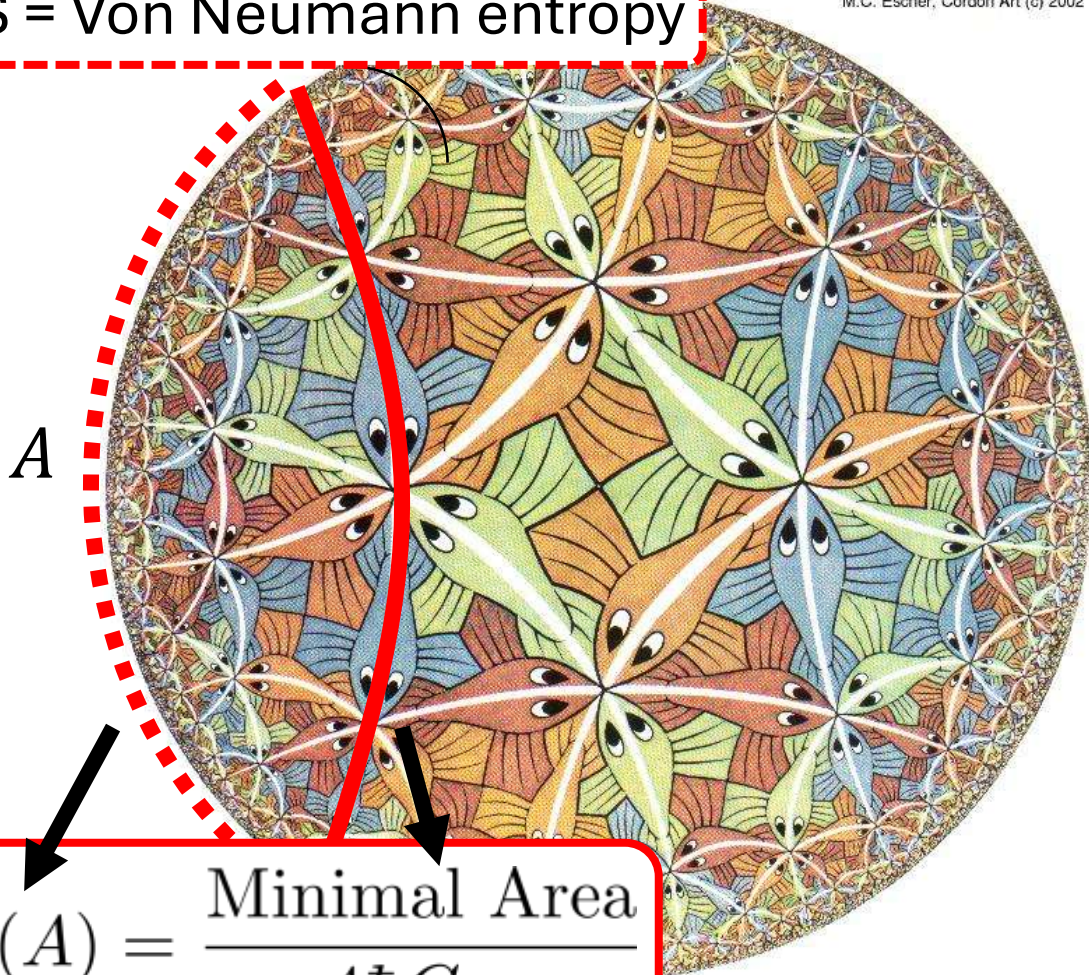
$$\xi = (x_1, \dots, x_N, p_1, \dots, p_N) \quad V_{ij} = \frac{1}{2} \langle \{ \xi_i - \langle \xi_i \rangle, \xi_j - \langle \xi_j \rangle \} \rangle$$

[literature: continuous variable graph states, methods to compute entropies]

Example 1: hyperbolic disk

S = Von Neumann entropy

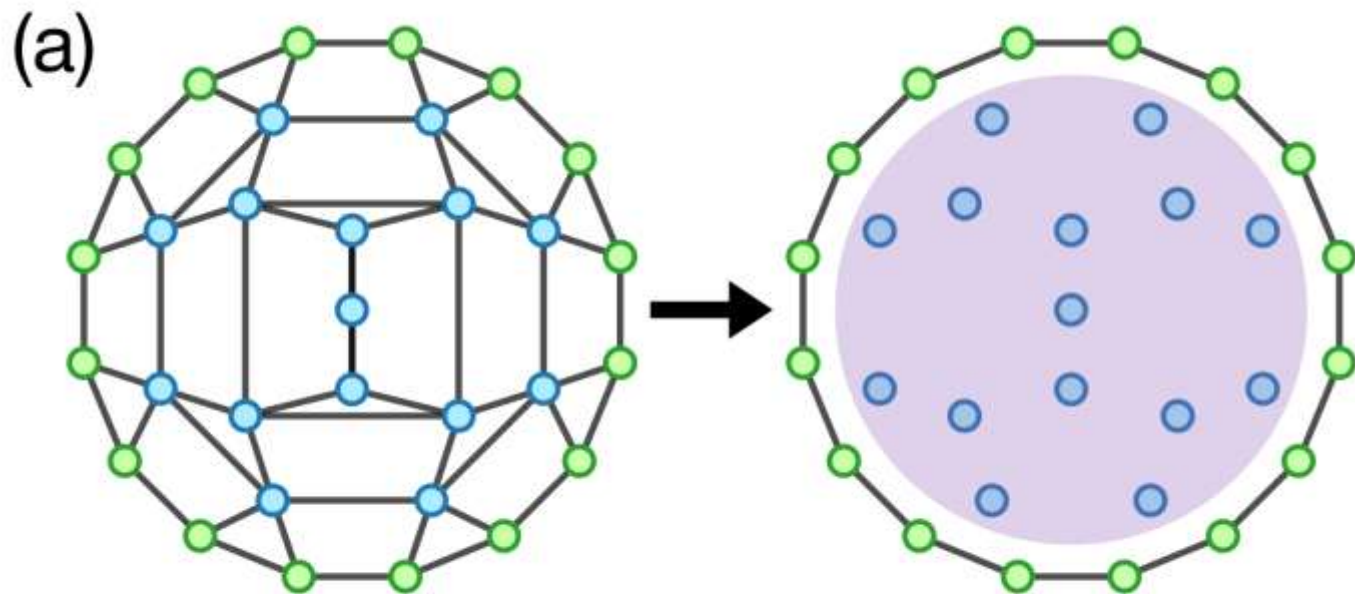
M.C. Escher, Gouda Art (c) 2002



- **AdS/CFT**: the ground state of the boundary (1+1)d CFT is dual to empty (2+1)d AdS space, spatial geometry is a hyperbolic disk
- The RT surface (curve in 2+1) is a portion of a circle (set $\hbar = 1$)

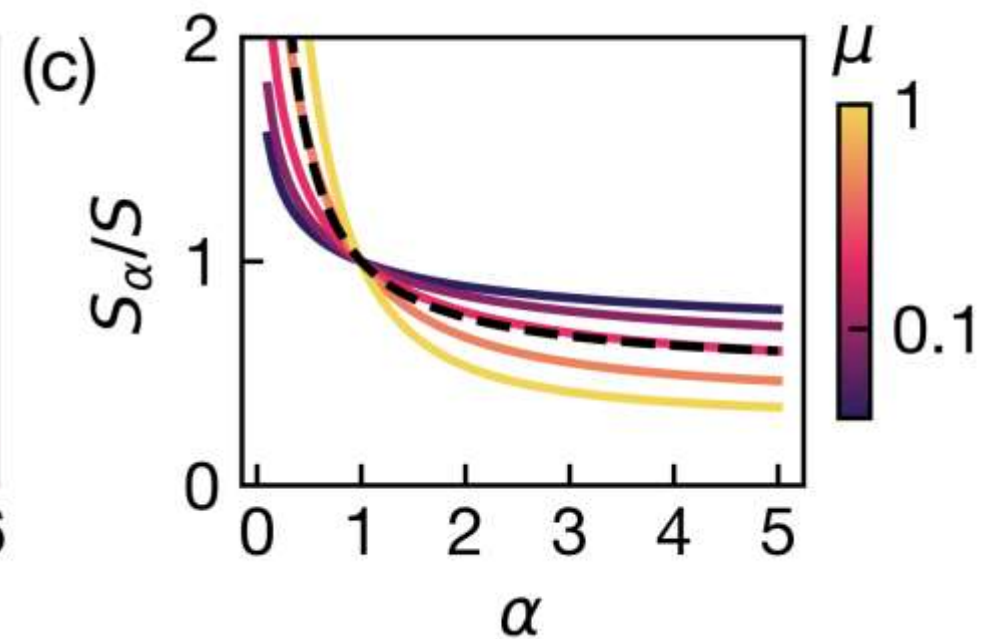
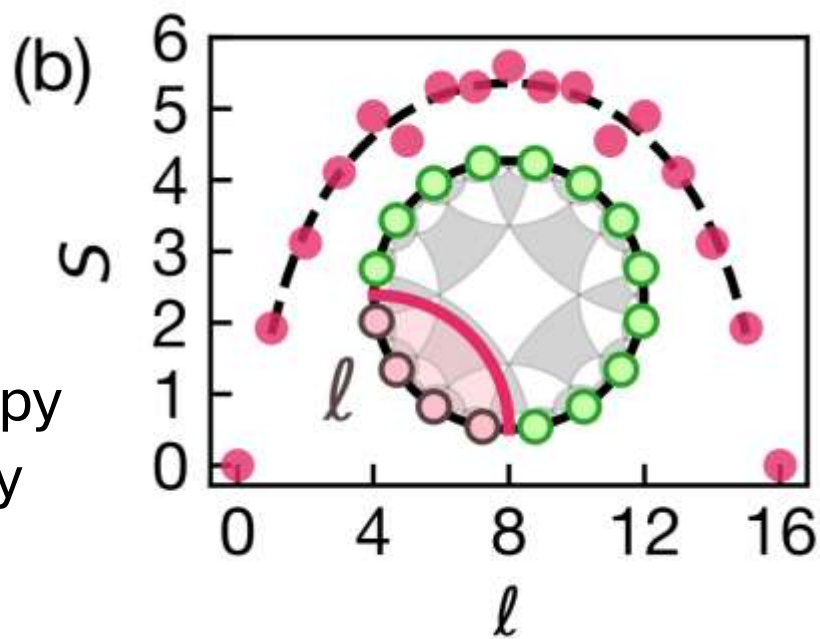
$$S = \frac{d_{\text{bulk}}}{4G} = \frac{\ell_{\text{AdS}}}{2G} \ln \left(\frac{2r_c}{\ell_{\text{AdS}}} \sin \frac{\Delta\theta}{2} \right)$$

$$S(A) = \frac{\text{Minimal Area}}{4\hbar G_N}$$



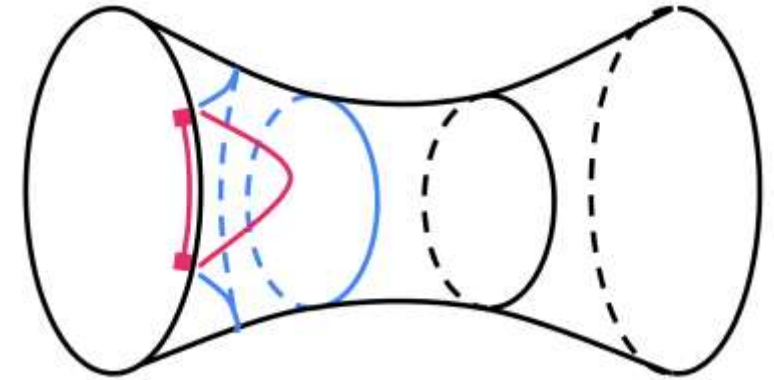
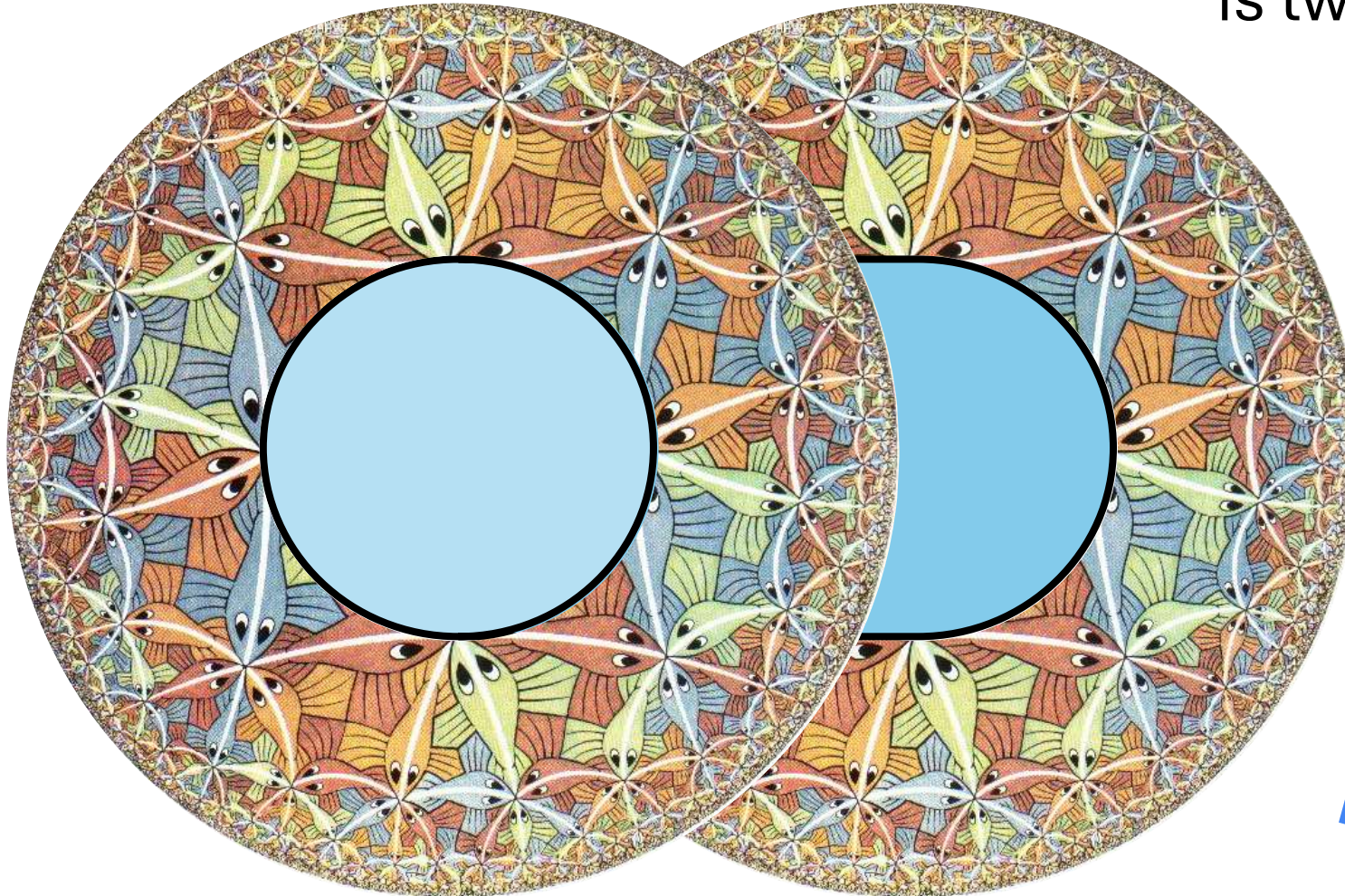
A simple discretization of the hyperbolic disk, followed by measurement of the bulk sites

(b) Entanglement entropy $\mu = .2$, (c) Renyi entropy as a function of μ



Example 2: wormhole

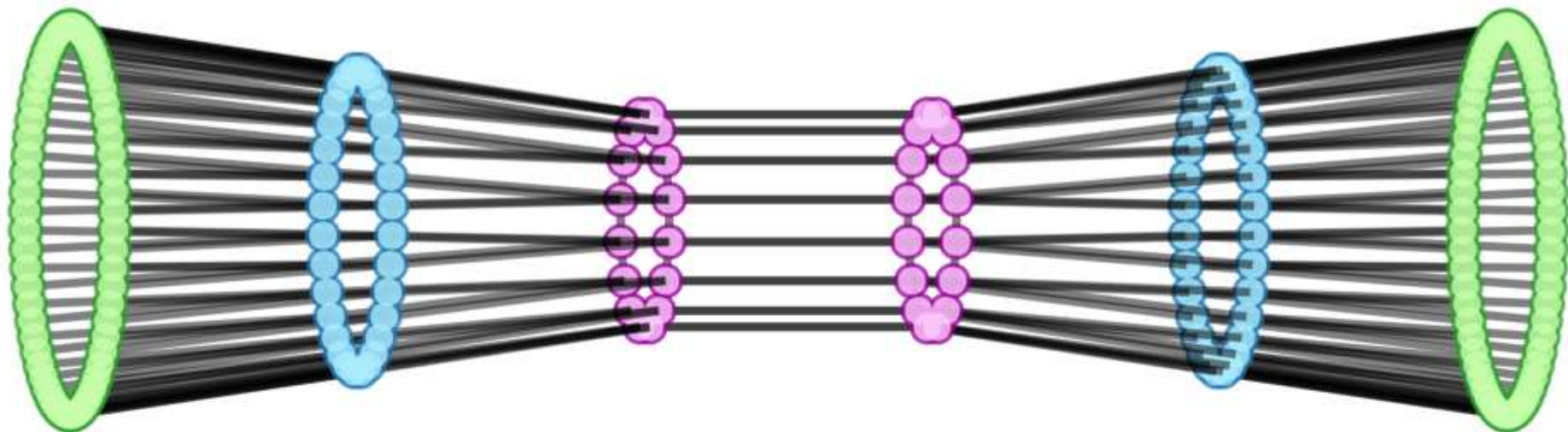
- **AdS/CFT**: a thermofield double state of the boundary (1+1)d CFT is dual to a wormhole connecting two black holes, spatial geometry is two annuli connected by a tube



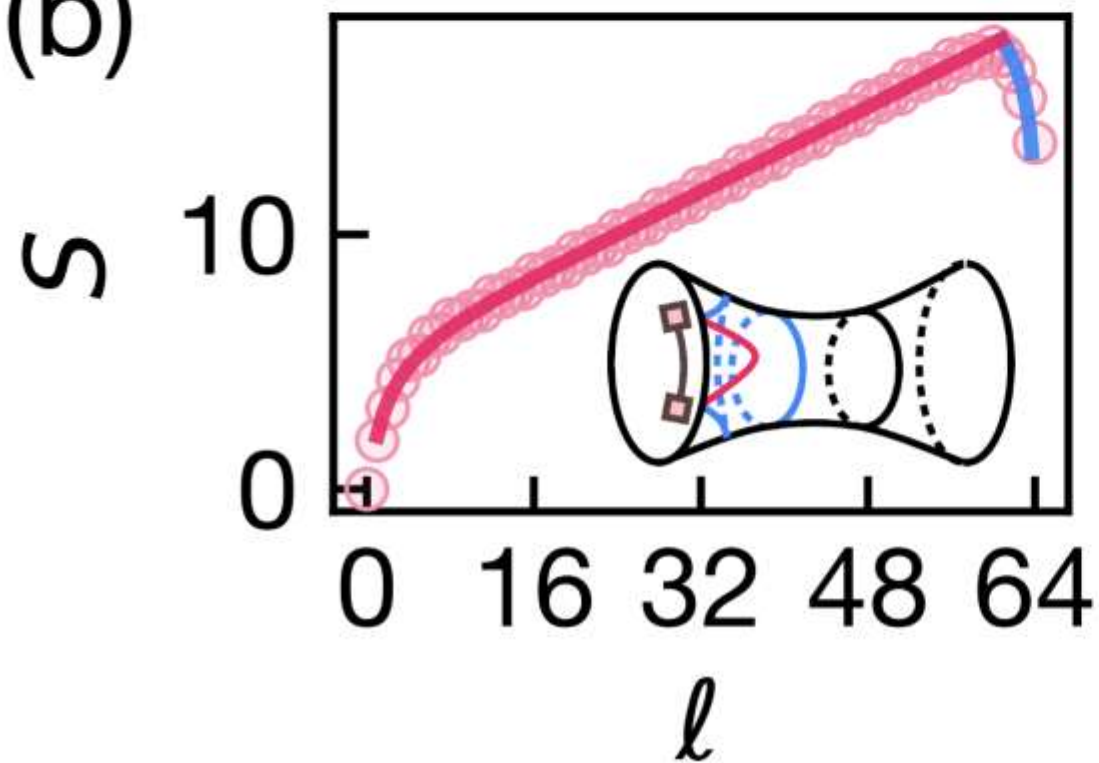
/
$$S^{(1)} = \frac{d_{\text{bulk}}}{4G} = \frac{c}{3} \ln \left(\frac{2r_c}{r_+} \sinh \frac{r_+ \Delta\theta}{2\ell_{\text{AdS}}} \right)$$

/
$$S^{(2)} = S_{th} + \frac{c}{3} \ln \left(\frac{2r_c}{r_+} \sinh \frac{r_+ (2\pi - \Delta\theta)}{2\ell_{\text{AdS}}} \right)$$

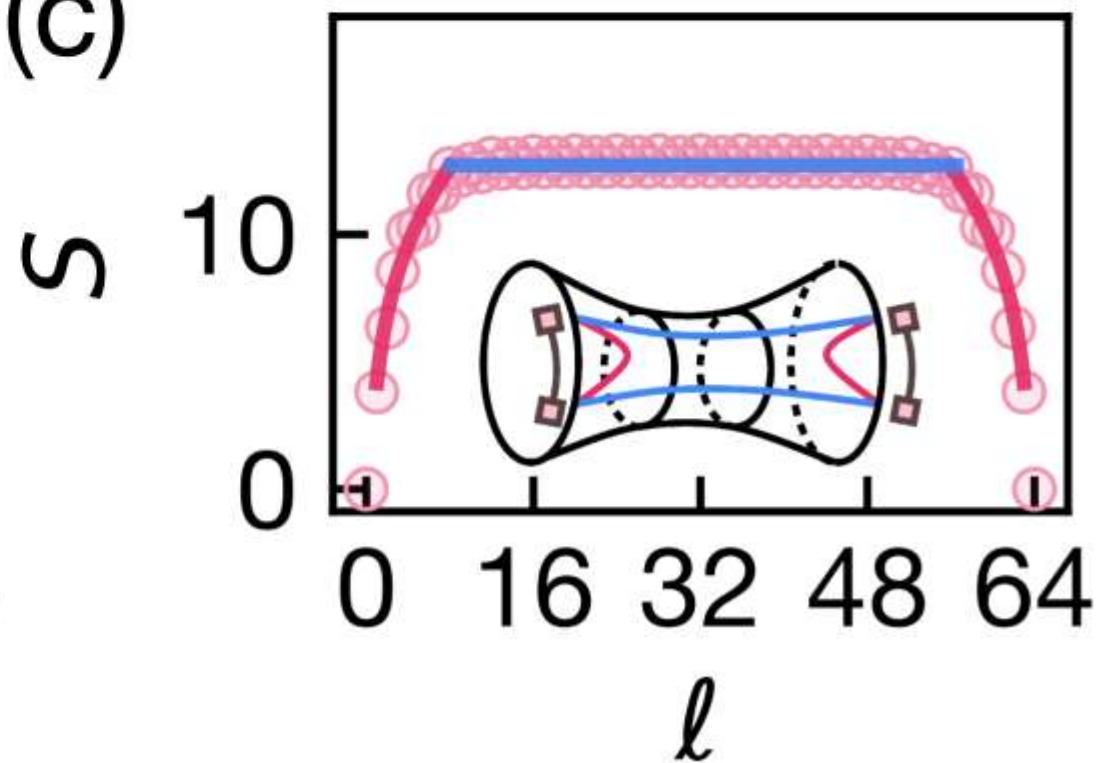
(a)



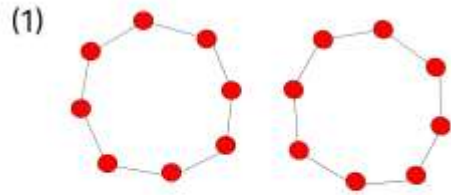
(b)



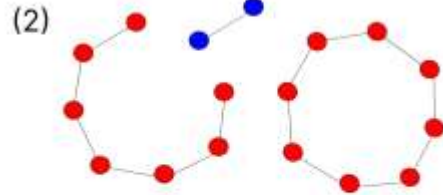
(c)



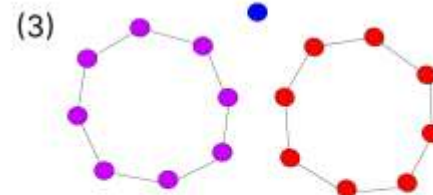
Wormhole-like teleportation?



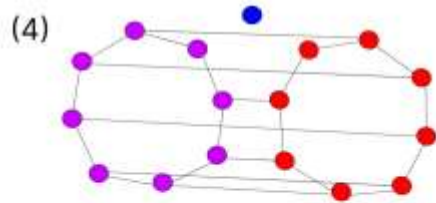
Start with TFD state and evolve left backwards in time



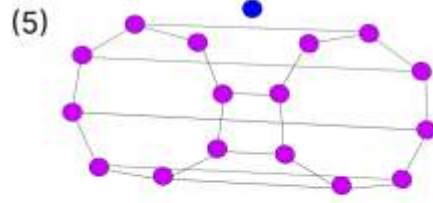
Replace one mode on left with a TMS state



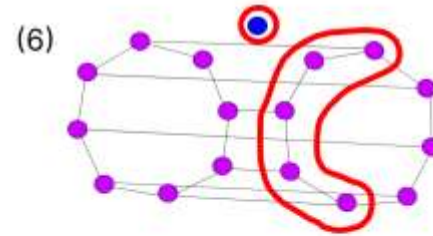
Evolve left forwards in time



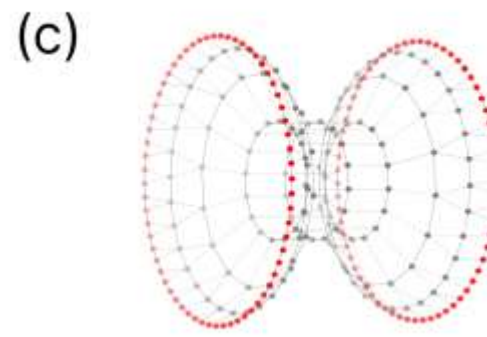
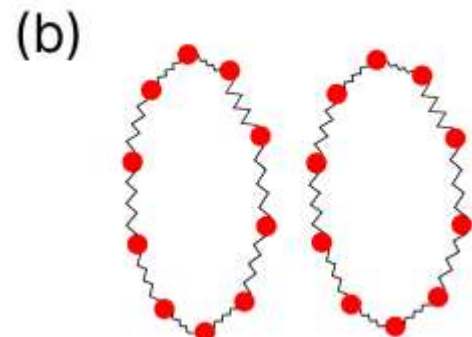
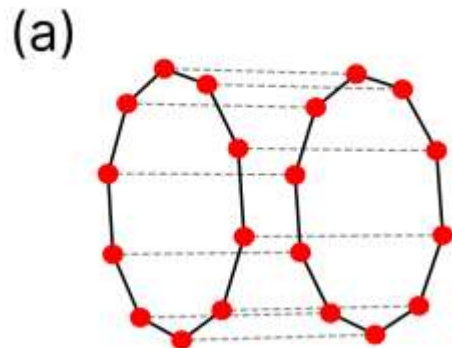
Couple the two sides excluding insert site and teleport site



Evolve right forwards in time



Compute mutual information between observer and segments



Allison Powell

Making thermal states

Preparing High Fidelity Thermofield Double States



Brian Khor



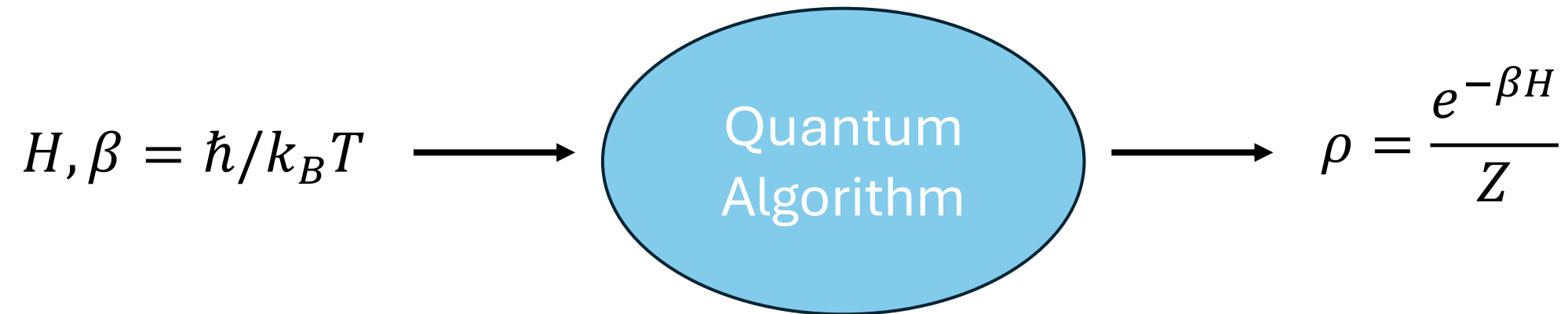
Nadie Li



Martin Sasieta

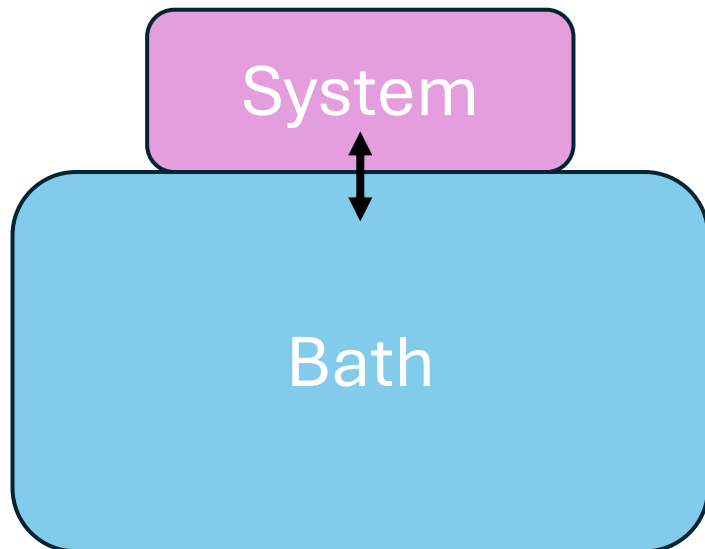
Preparing thermal states

- For all kinds of reasons, ranging from basic science to applied materials research, we naturally want to have **general purpose algorithms** to efficiently prepare thermal states



Is this even possible?

- Nature provides us with one relatively general algorithm: couple the system of interest to a heat bath at the desired temperature and let the system come into equilibrium with the bath
- This is general purpose but it may take a long time to equilibrate*



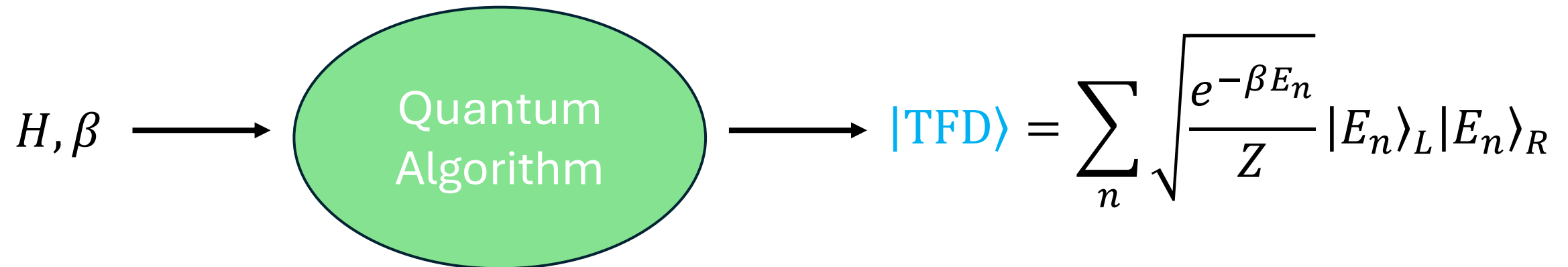
$$\rho_S(t \rightarrow \infty) \approx e^{-\beta H} / Z$$

*This is a real limitation, for example:

- Glasses may not equilibrate on any reasonable timescale
- Critical slowing down at a continuous phase transition
- Encoding of hard optimization problems suggests a barrier

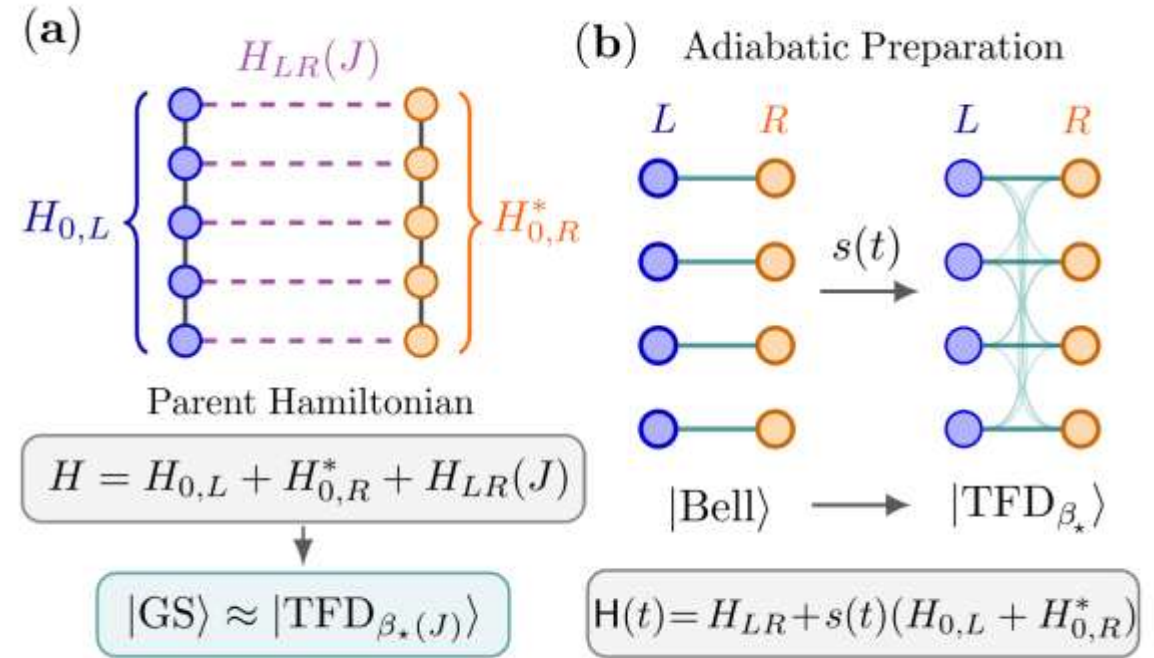
Thermofield double states

- Sometimes we want something stronger: a special structured purification of the thermal state, a **thermofield double state**
- This state shows up in quantum gravity, field theory, and many-body physics in various important ways
- **Nature's algorithm does NOT provide this**



Our protocol

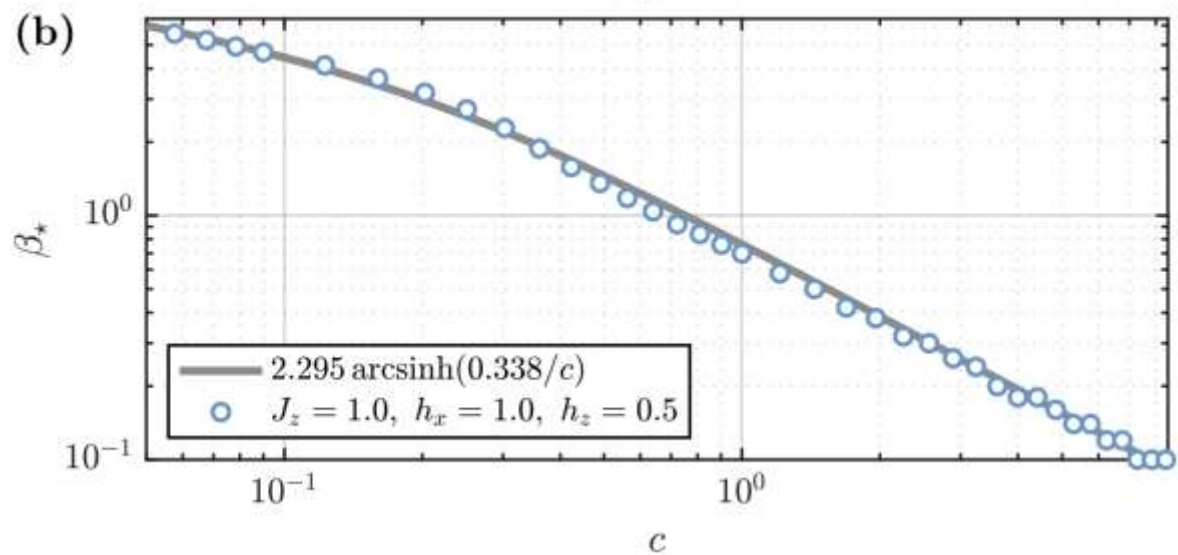
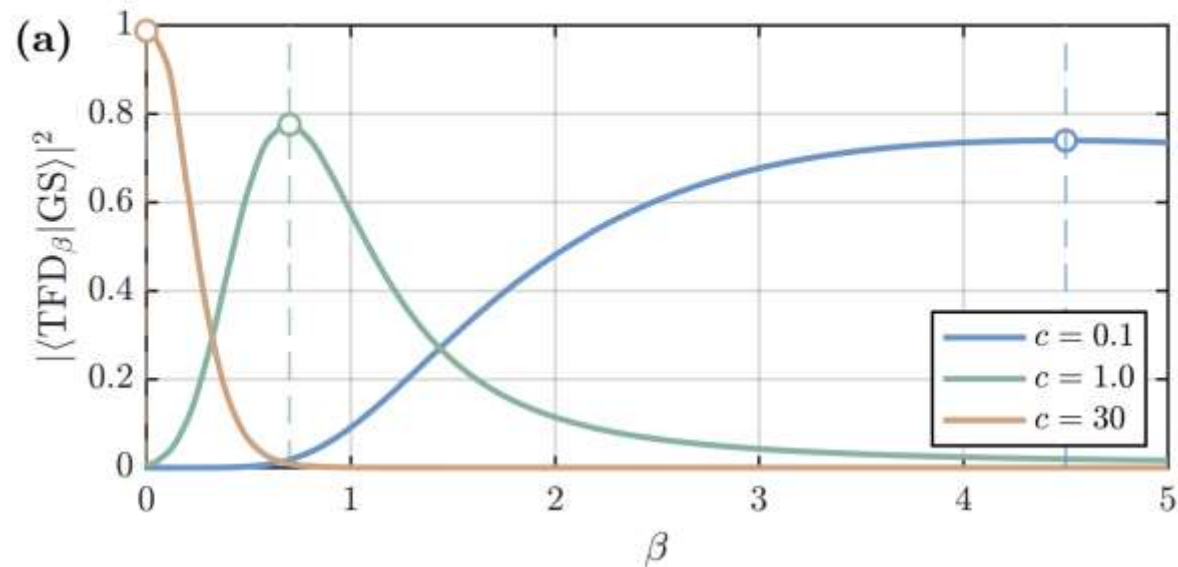
$$H = H_{0,L} + H_{0,R}^* + \frac{JN}{2K} \sum_{\alpha=1}^K (\mathcal{O}_{\alpha,L} - \mathcal{O}_{\alpha,R}^*)^2,$$



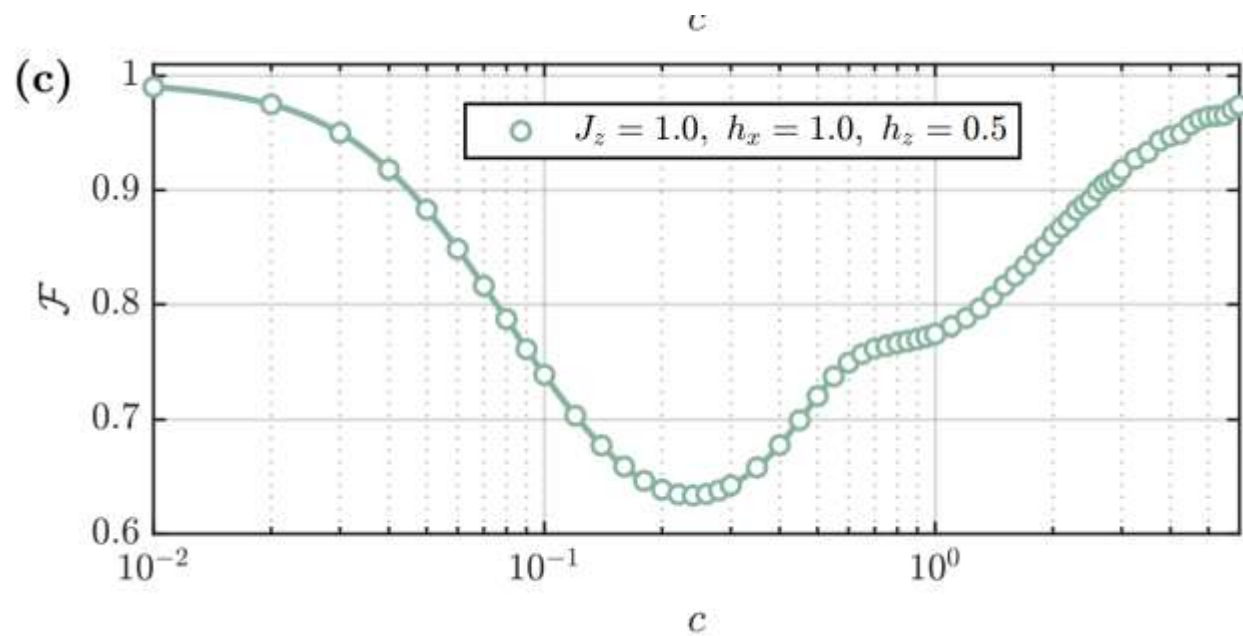
- Part 1: Construct a Hamiltonian family whose GS is an approximate TFD at a tunable temp (**controlled by LR coupling**), adiabatic preparation
- Part 2: Improved fidelity by adding a penalty term that favors a diagonal subspace, $|E_n\rangle_L |E_n\rangle_R$

We provide evidence that this will work when the system self-thermalizes

Mixed Field Ising ($N = 30$)

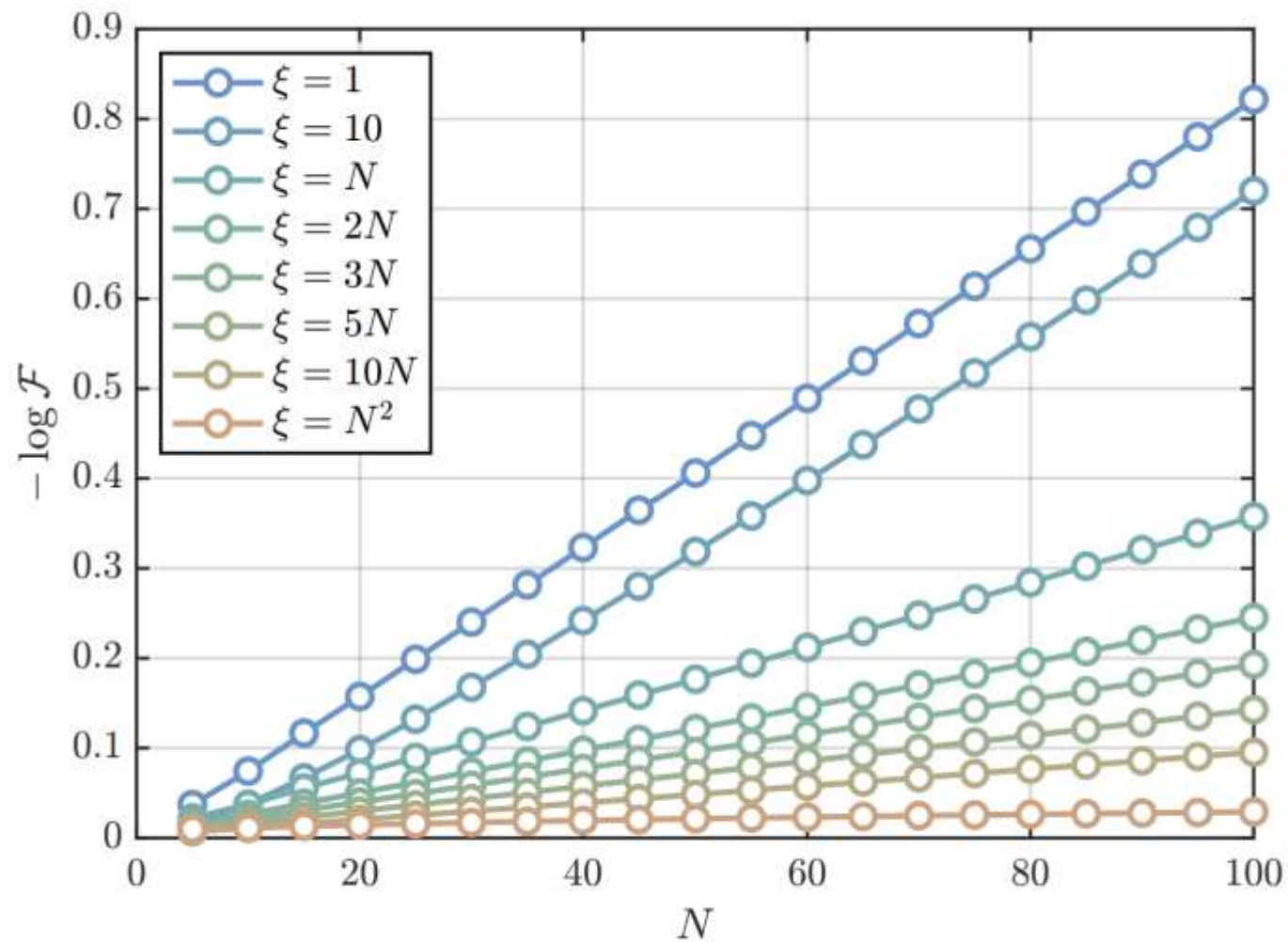


$$H_{\text{MFI},L/R} = J_z \sum_{i=1}^{N-1} Z_{i,L/R} Z_{i+1,L/R} + h_x \sum_{i=1}^N X_{i,L/R} + h_z \sum_{i=1}^N Z_{i,L/R}, \quad (11)$$



Penalty term

$$\frac{\xi}{2N} (H_{0,L} - H_{0,R}^*)^2,$$



Gravity calculation + general argument

- For the SYK model in its low-energy regime, we can do some further analytics to get (e.g. $\Delta_{\mathcal{O}} = 1/4$):

$$\log \mathcal{F} |_{\xi} \sim -N \times (\text{small coefficient}) \times (N/\xi)^{2-4\Delta_{\mathcal{O}}},$$

$$\mathcal{F} \xi \gtrsim N^{\frac{3-4\Delta_{\mathcal{O}}}{2-4\Delta_{\mathcal{O}}}}.$$

- A general perturbative calculation suggests $\xi \gtrsim N^3$ may be sufficient to stick in the diagonal subspace

$$\|P_{\text{off}}|\text{GS}\rangle\|^2 \sim \frac{N^2}{K} \int d\omega \frac{\hat{\rho}_{\text{off}}(\omega)}{(\Delta_{\text{off}} + \xi\omega^2/N)^2},$$

Measuring scrambling

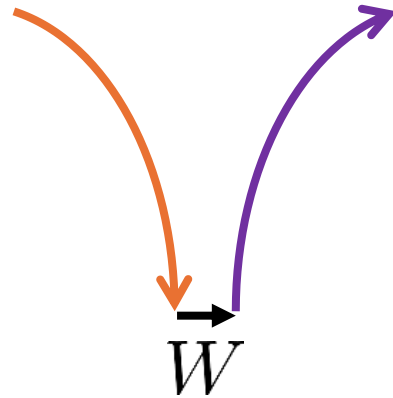
The Butterfly Effect.



by
J.L. Westover

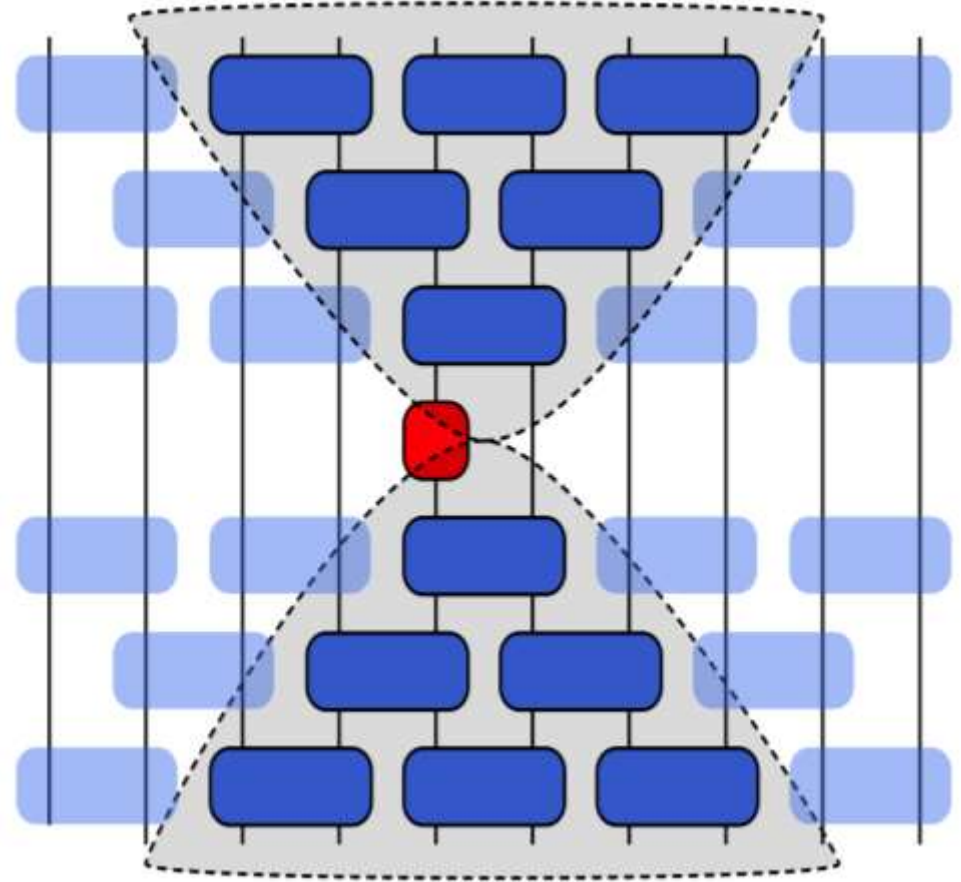
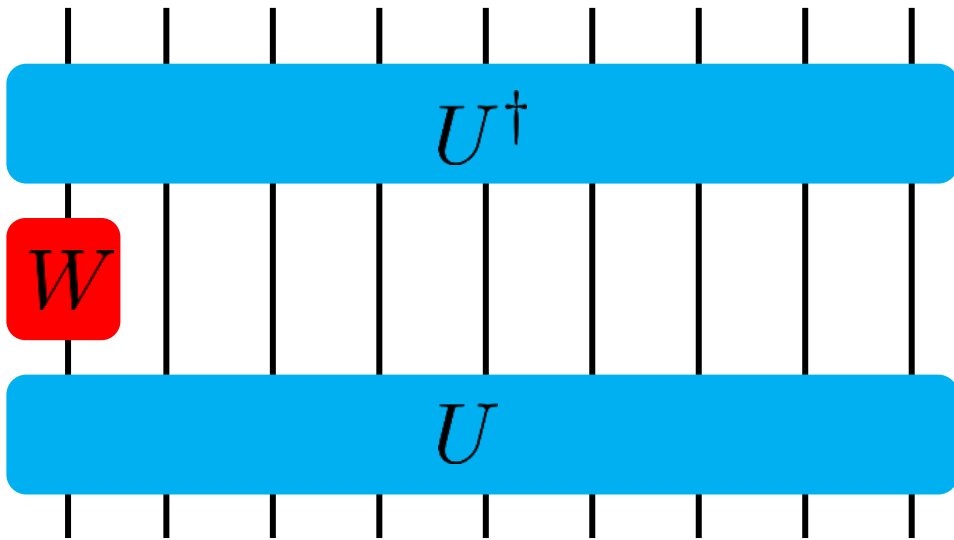
www.mrlovenstein.com

$$W(t) = \underbrace{e^{iHt/\hbar}}_{\text{backward}} \underbrace{W}_{\text{perturbation}} \underbrace{e^{-iHt/\hbar}}_{\text{forward}}$$



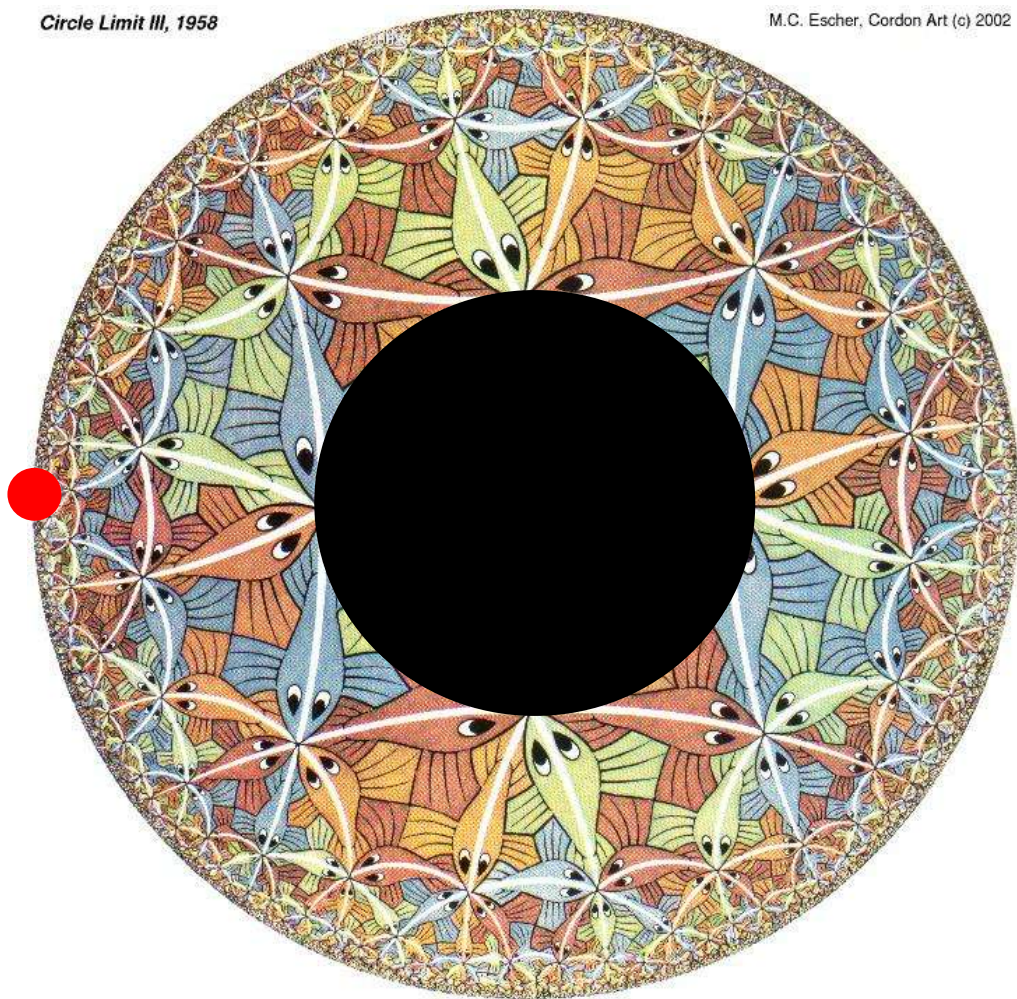
Heisenberg operator growth

$$W(t) = U^\dagger W U$$



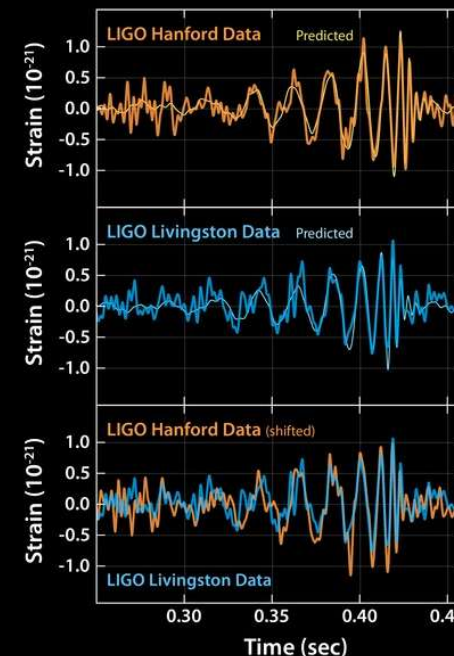
Circle Limit III, 1958

M.C. Escher, Cordon Art (c) 2002



Perturbation grows as it falls

$$\tau \sim \frac{\hbar}{T}$$



QG: scrambling arises from an exponential growth of energy as a particle falls towards a black hole, $E \sim e^{2\pi t/\beta} \sim e^{\lambda t}$

[Shenker-Stanford]

λ is like a Lyapunov exponent



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Measuring the Scrambling of Quantum Information



Greg Bentsen



Monika Schleier-Smith



Patrick Hayden

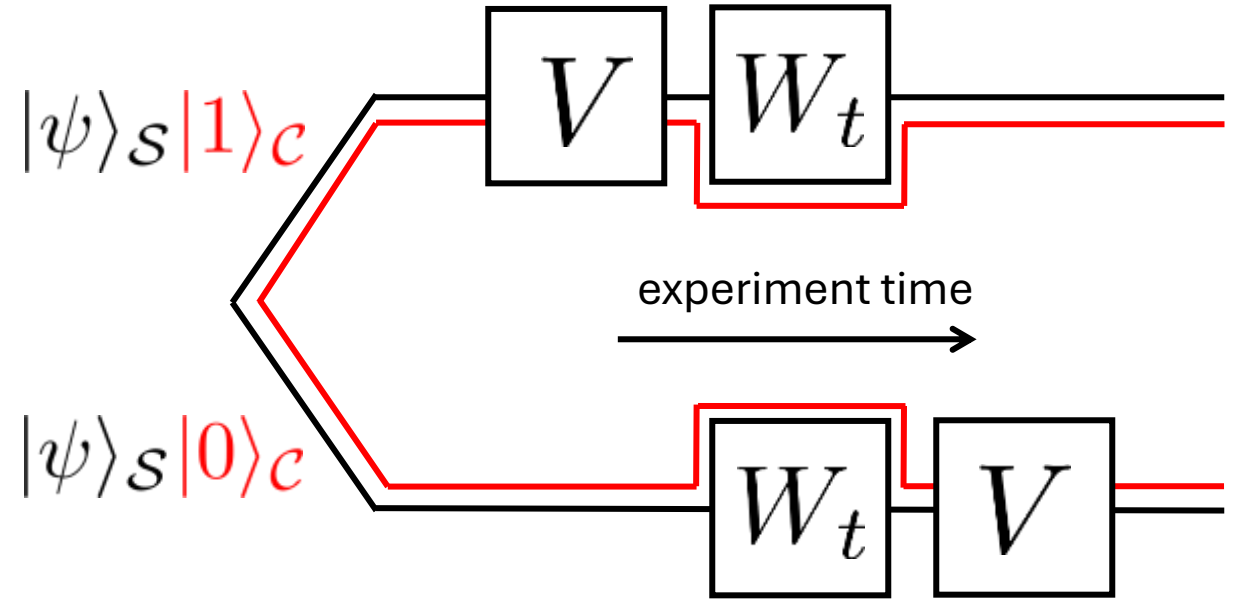
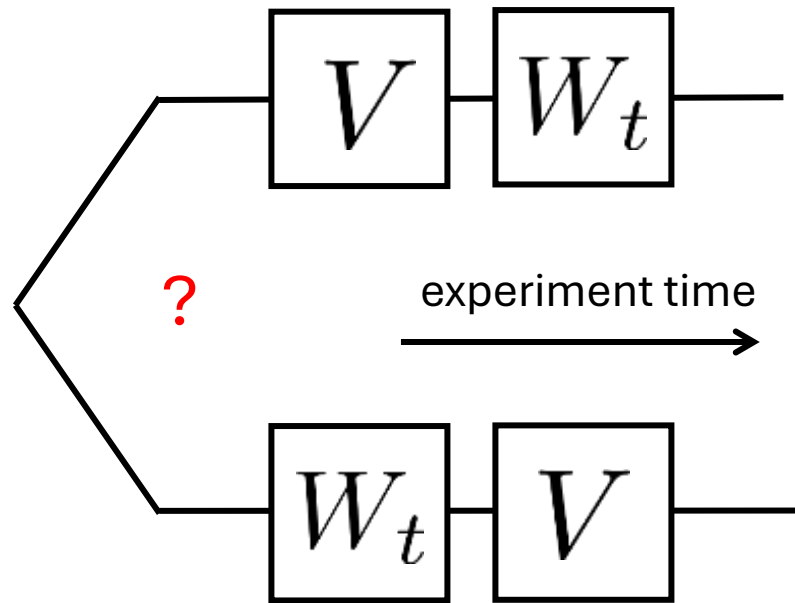
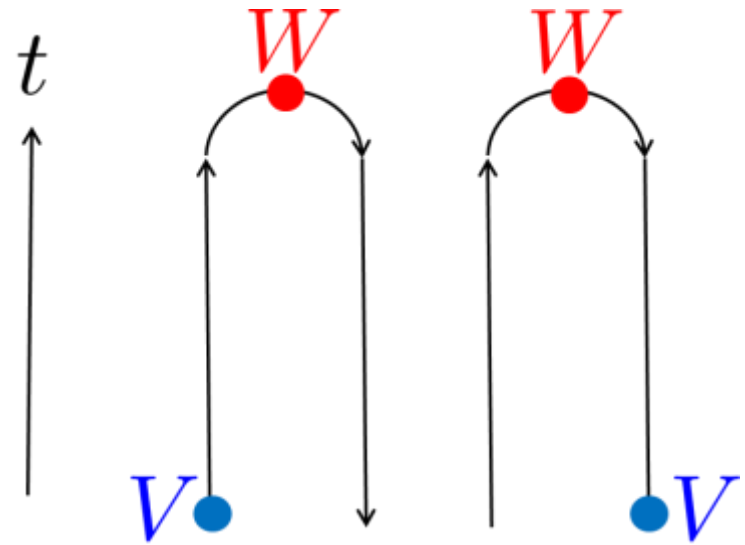
***Physical Review A* 50th Anniversary
Milestones**

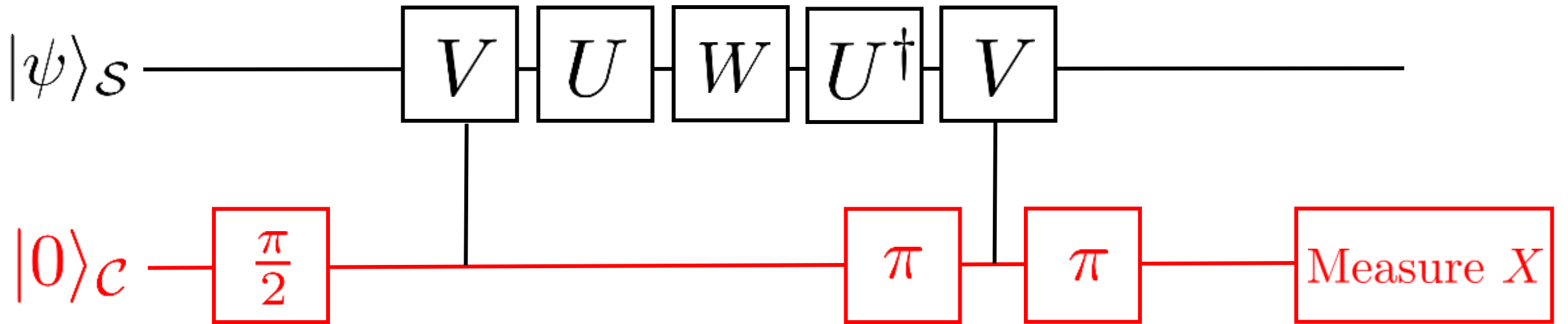


PHYSICAL
REVIEW A

Interferometric Protocol

$$F = \left[\langle \psi | W_t^\dagger V^\dagger \right] \left[W_t V | \psi \rangle \right]$$

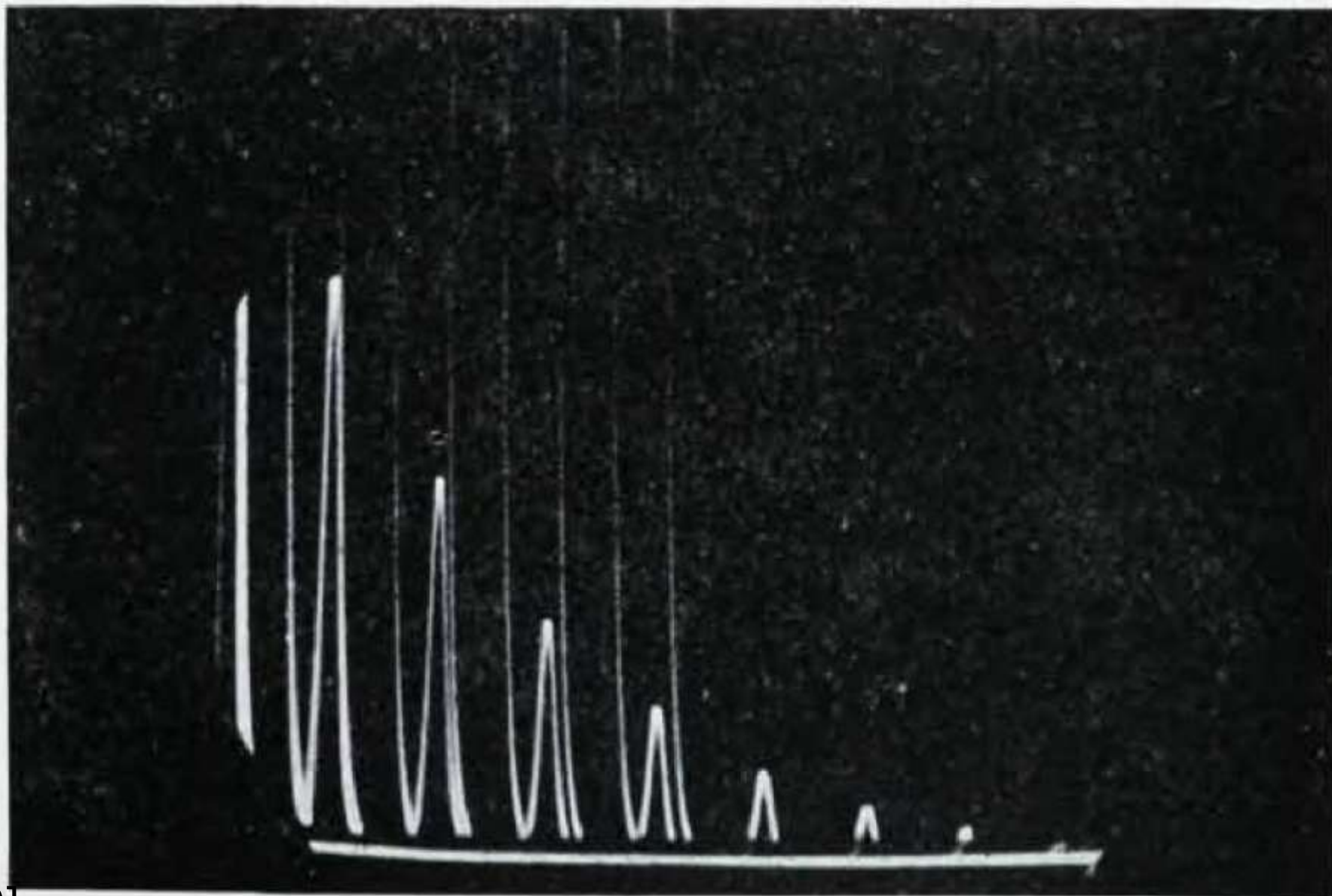




Final state:
$$\frac{VW_t|\psi\rangle_S|0\rangle_C + W_tV|\psi\rangle_S|1\rangle_C}{\sqrt{2}}$$

Measure Pauli X of the control: $\langle X_C \rangle = \text{Re}[F]$

Fig. 6. Multiple exposures of single proton echoes. The first rf pulse occurs at the beginning of the trace and the second pulse is spaced from the origin at equal intervals for each exposure with the sample at thermal equilibrium. The echo envelope provides a measure of the phase coherence parameter T_2 .

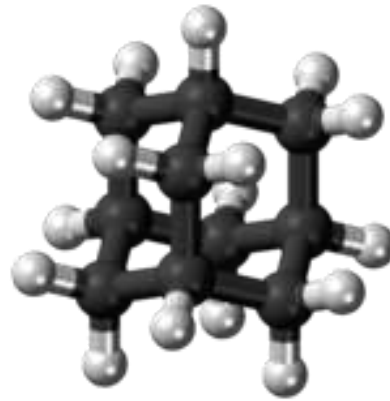


[Spin Echoes: Hahn 1950]

Measuring Global OTOCs in adamantane

Solid adamantane is a classic NMR system: **an fcc lattice of molecules that tumble in place**

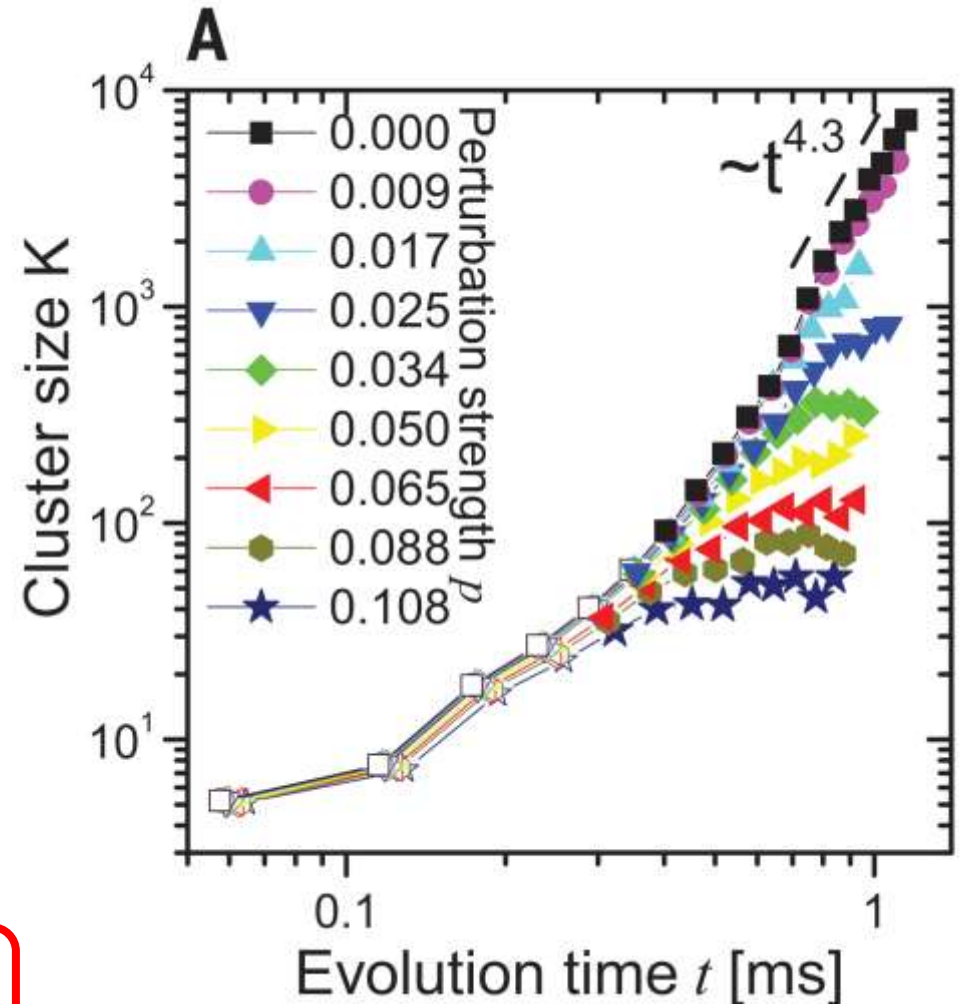
[my intro via: [Pastawski-Levstein-...](#)]



(p=0 case)

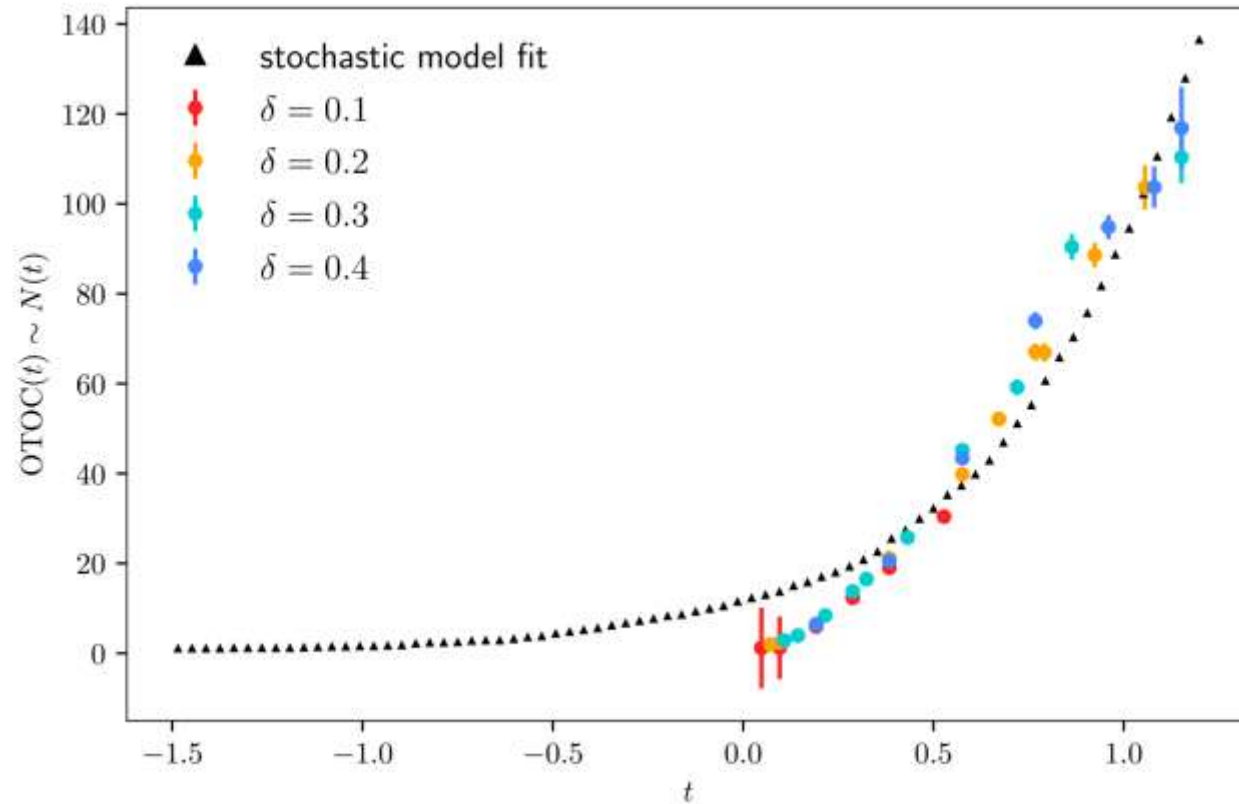
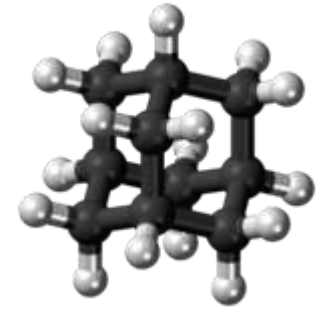
$$C_{\text{global}} = -\langle [I_{\text{total}}^z(t), I_{\text{total}}^z(0)]^2 \rangle_{T=\infty}$$

Can we understand these measurements?

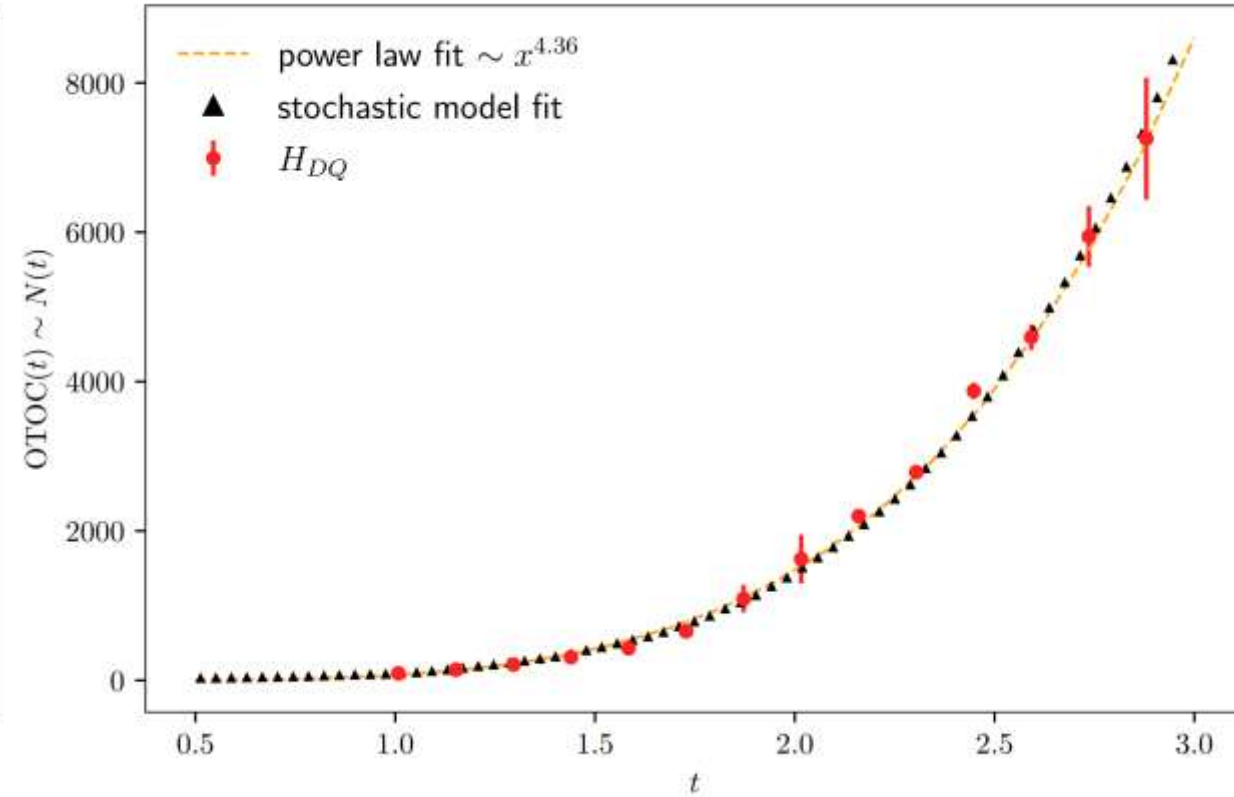


[[Alvarez-Suter-Kaiser Science '15](#)]

Model of adamantane: $N=16$, fcc lattice



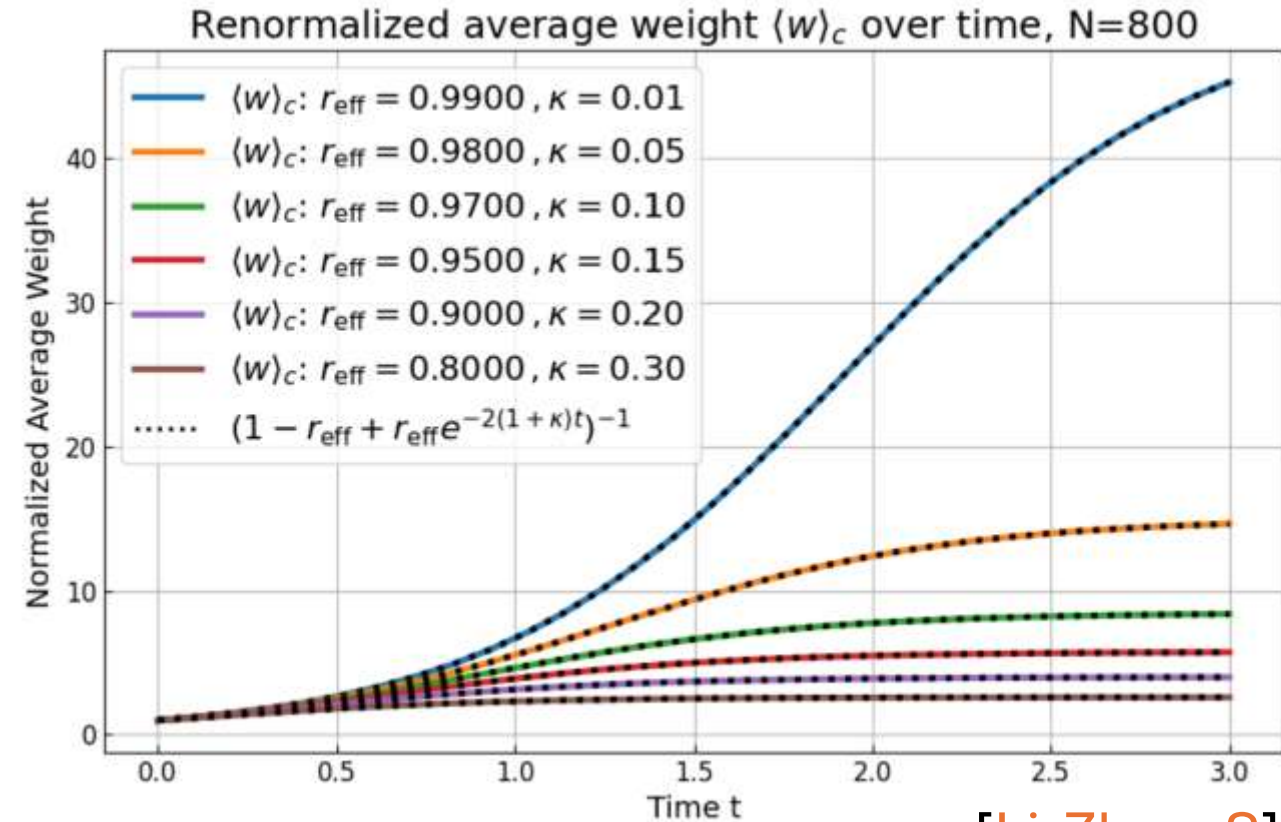
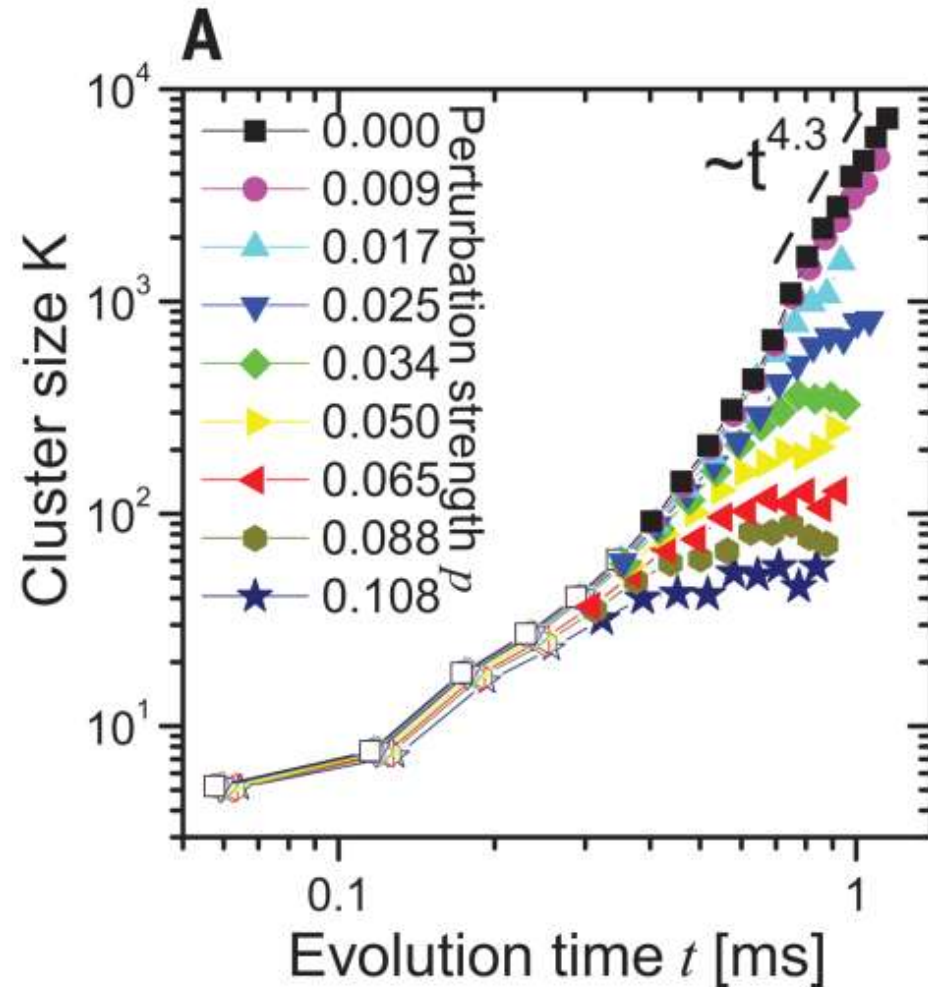
[Y-type dipole data: Sanchez et al. PRL 124, 030601 (2020), model: Zhou-S]



[double quantum data: Alvarez-Suter-Kaiser Science 2015, model: Zhou-S]



Perturbations in a solvable model



Towards cosmology



Valerie Bettaque



Shiyong Guo



Alejandro Vilar Lopez

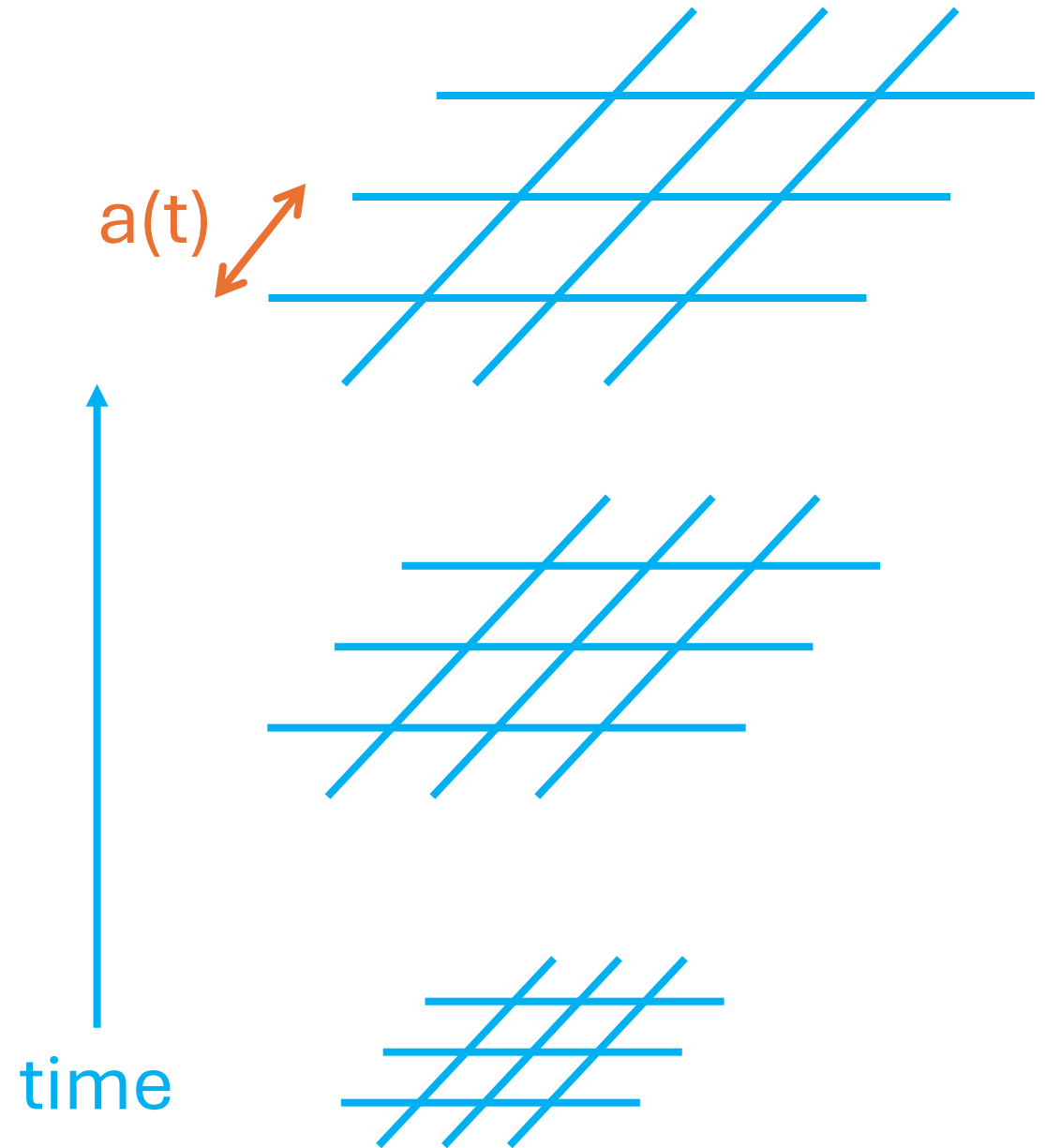
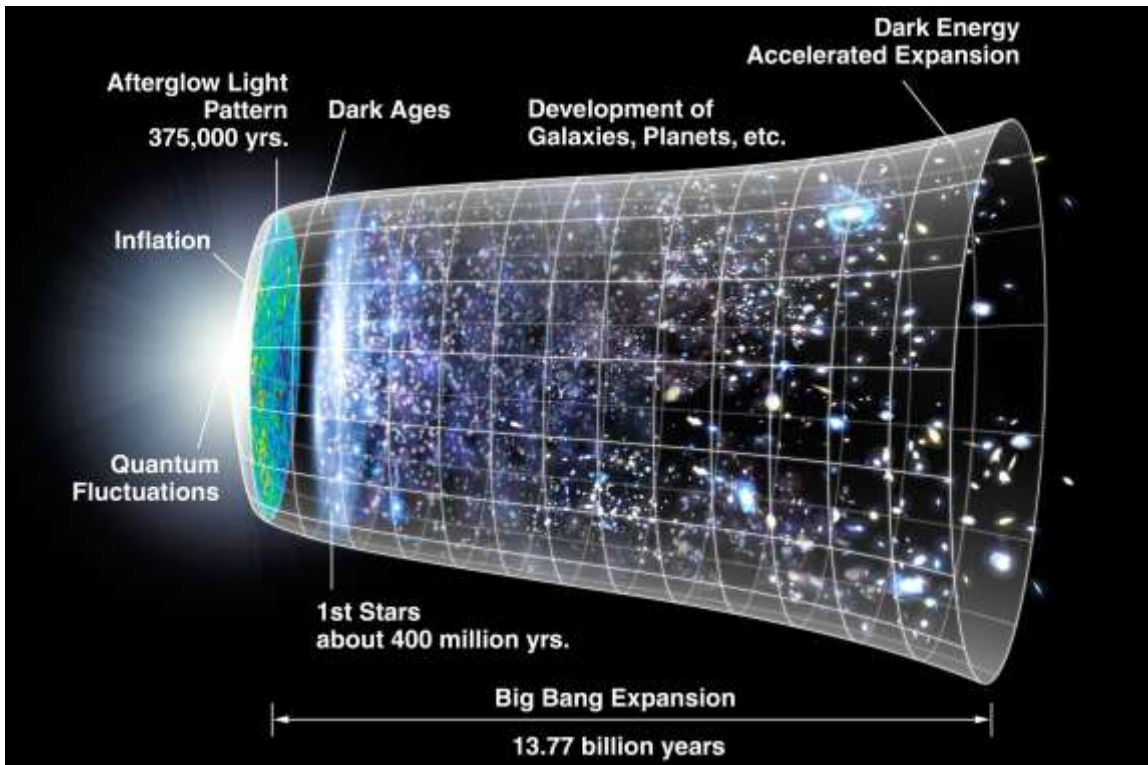


Martin Sasieta

+ AI systems (Opus,
Fable, Sol, Astra)

But our universe is:

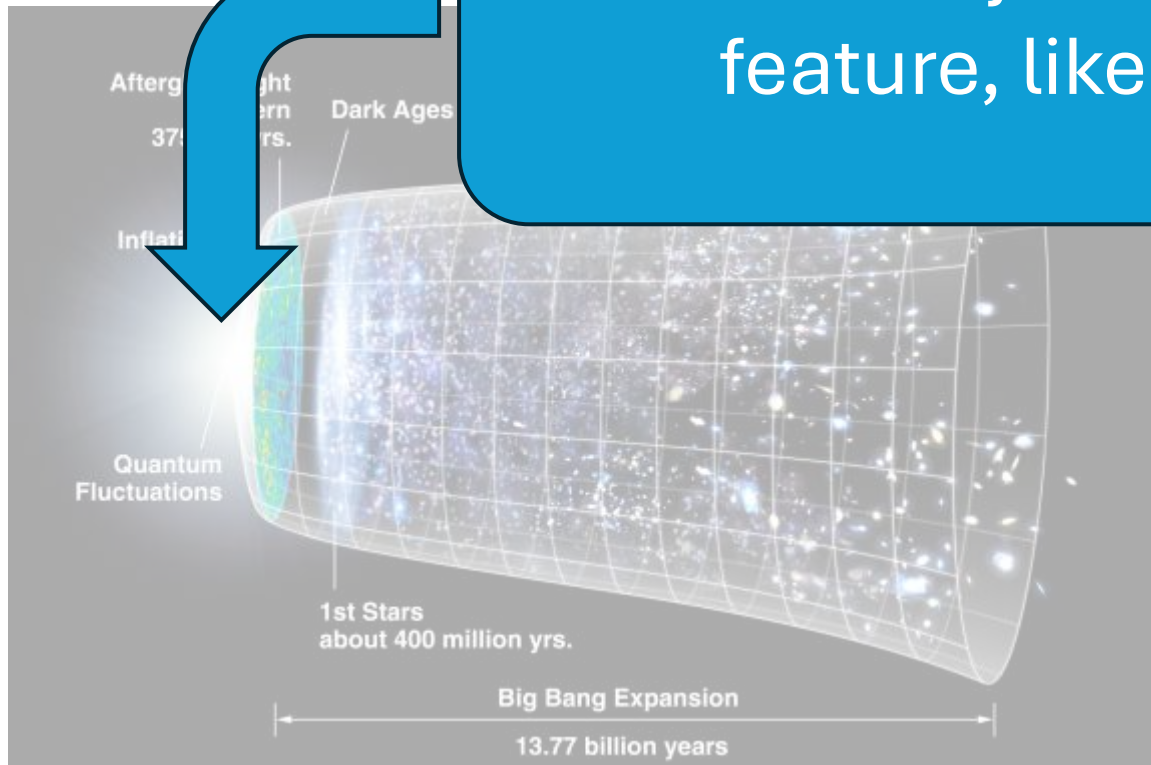
- Full of stuff, homogenous, ...
- Expanding: $a(t)$
- Λ CDM: positive dark energy
- **No AdS boundaries**



But our universe is:

- Full of stuff, homogenous, ...
- Expanding: $a(t)$
- Λ CDM: positive dark energy
- No AdS background

What if we just wanted a single feature, like a Big Bang?



time

Consider a system of N Majorana fermions χ_i

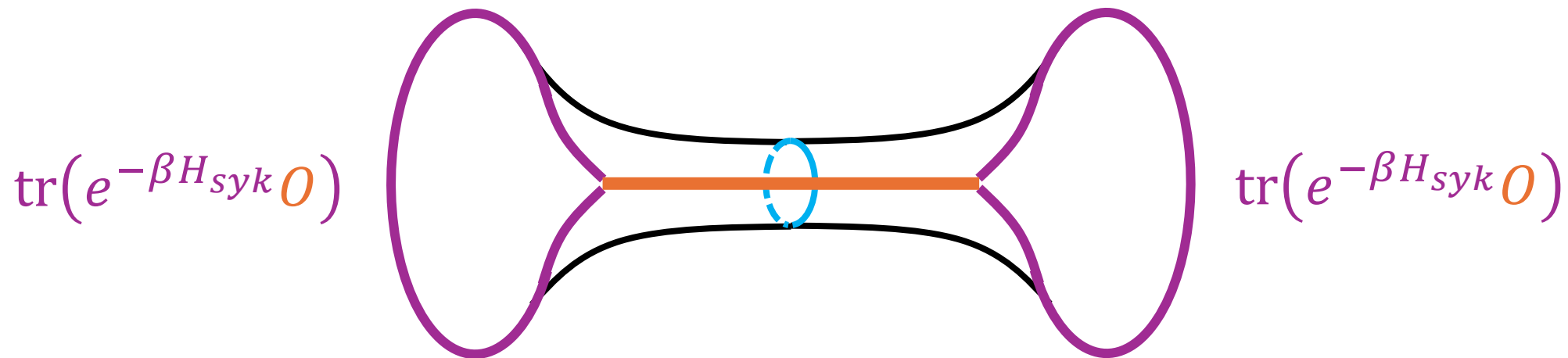
Majorana strings, a complete basis of Hermitian operators: $\mu(a) \propto \chi_1^{a_1} \cdots \chi_N^{a_N}$ Weight = $\sum a_i$

Thermal 1-point functions, these completely determine the state: $\xi(a) = \frac{\text{tr}(e^{-\beta H} \mu(a))}{\text{tr}(e^{-\beta H})}$

Surprisingly, statistical properties of these thermal 1-point functions will have something to do with cosmology ...

QG geometrizes the statistical variance

- A Euclidean wormhole gives a contribution to a statistical second moment (two boundaries) of erratic data in the boundary theory
- We will make this precise using the Sachdev-Ye-Kitaev model



[connection to averaging: Coleman, Saad-Shenker-Stanford (SFF), Antonini-Waddell-Simidzija-Van Raamsdonk-S (cosmological), Stanford (heavy op), ...]

Sachdev-Ye-Kitaev model ($q=4$)

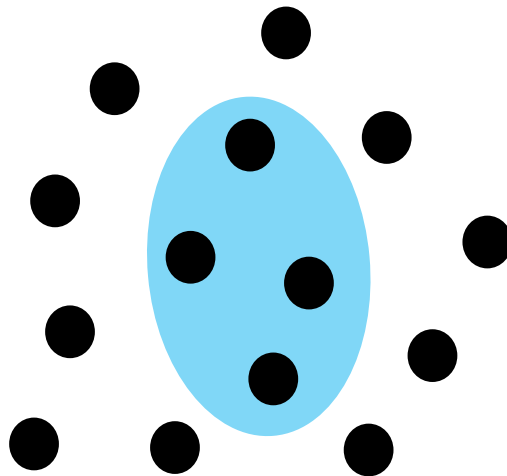
$$\chi_a^\dagger = \chi_a, \quad a = 1, \dots, N$$

$$H = \frac{1}{4!} \sum_{abcd} J_{abcd} \chi_a \chi_b \chi_c \chi_d$$

$$\{\chi_a, \chi_b\} = \delta_{a,b}$$

$$\overline{J_{abcd}} = 0, \quad \overline{J_{abcd}^2} = \frac{3! J^2}{N^3}$$

$$D = 2^{N/2}$$



[random 2-body ensemble, Sachdev-Ye, Kitaev, Rosenhaus-Polchinski, Maldacena-Stanford, Garcia-Garcia-Verbaarschot, Bagrets-Altland-Kamenev, ...]

Bulk theory

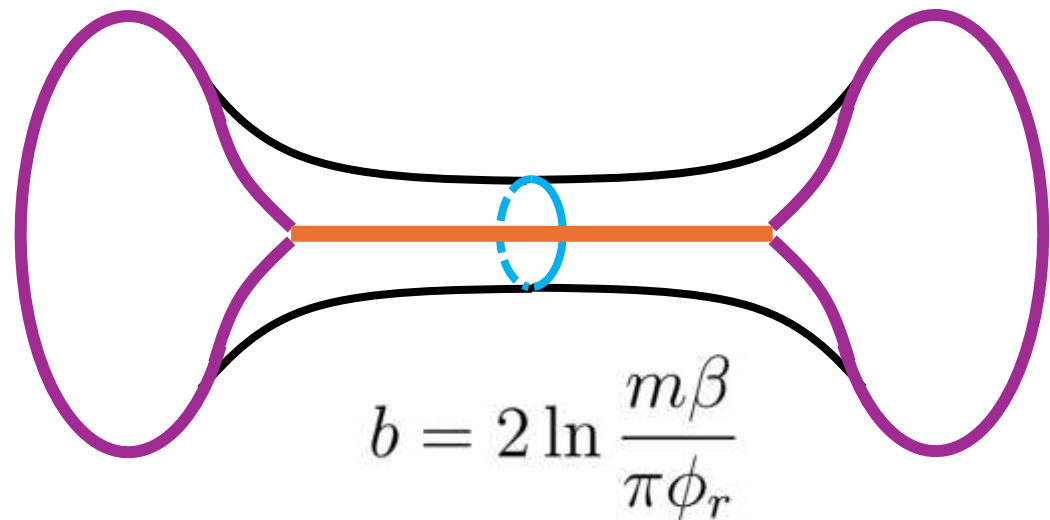
$$I_{\text{JT}} = -S_0\chi - \frac{1}{2} \int d^2x \sqrt{g} \Phi (R + 2) + (\text{boundary terms}) - m\ell_H + I_{\text{matter}}$$

- We model the SYK inter-replica correlation with a boundary-to-boundary correlations of N light probe fermions moving in a background geometry stabilized by a heavy particle [Bettaque-S]
- The bulk 2-point function of the light fermions is computed by summing over geodesics with mass $\rightarrow$ scaling dimension
- Background – double trumpet

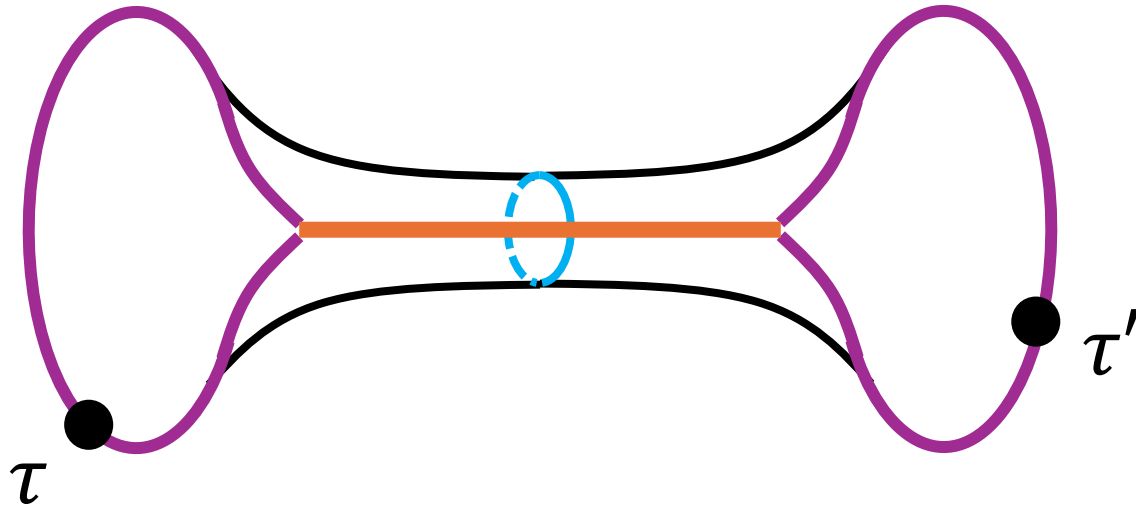
$$ds^2 = d\rho^2 + b^2 \cosh^2 \rho d\sigma^2$$

$$\Phi = \frac{\pi\phi_r}{\beta} \cosh \rho \cosh b\sigma.$$

[Stanford, Usatyuk-Wang-Zhao]



Wormhole calculation



$$e^{\rho_c} = \frac{2\beta}{\pi\epsilon} \frac{1}{\cosh b\sigma}.$$

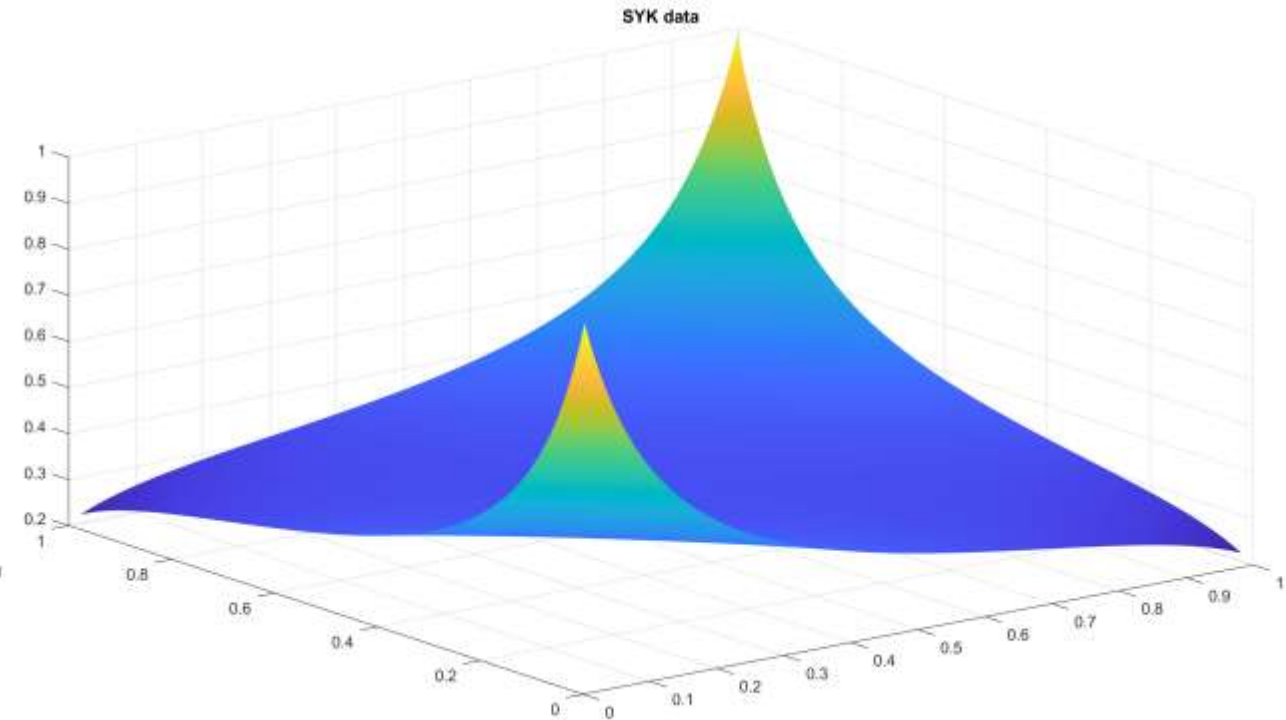
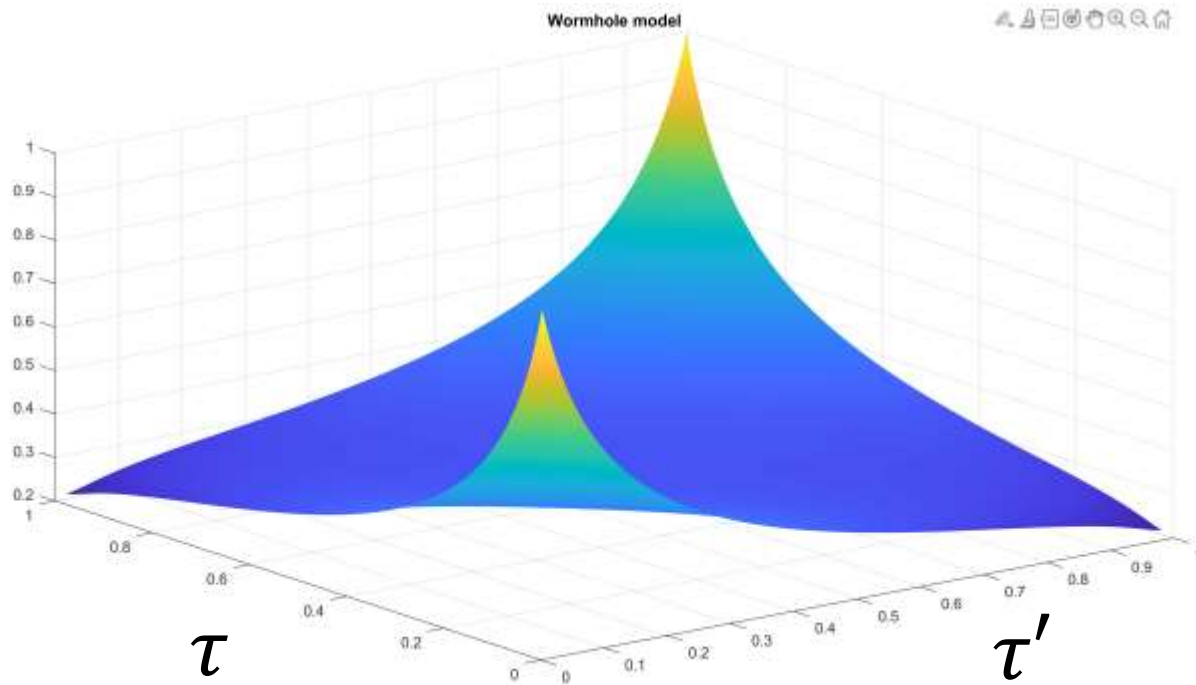
$$\frac{\tau(\sigma)}{\beta} = \frac{\arctan e^{b\sigma} - \arctan e^{-b/2}}{\arctan e^{b/2} - \arctan e^{-b/2}}.$$

winding geodesics $d_n(\sigma, \sigma') = \rho_c(\sigma) + \rho_c(\sigma') + \log \cosh^2 \frac{b(\sigma - \sigma' - n)}{2}.$

$$\mathcal{G}_P(\tau, \tau') = \mathcal{N}_P \sum_n e^{-\Delta d_n(\sigma(\tau), \sigma'(\tau'))}, \quad \mathcal{G}_{AP}(\tau, \tau') = \mathcal{N}_{AP} \sum_n (-1)^n e^{-\Delta d_n(\sigma(\tau), \sigma'(\tau'))},$$

Comparison of normalized correlations

$$\mathcal{G}_f = (1 - f)\mathcal{G}_{AP} + f\mathcal{G}_P.$$

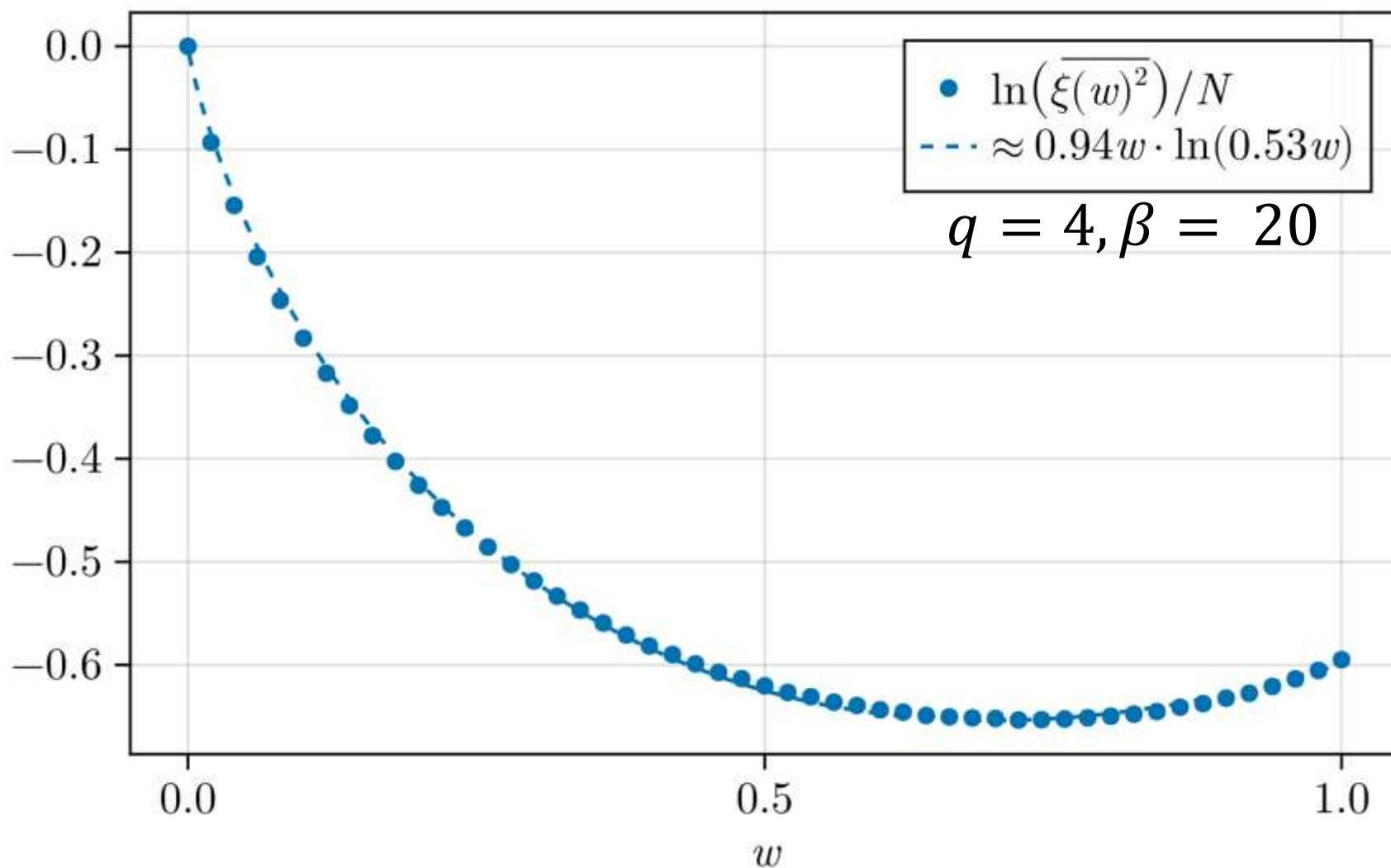


Bulk model: $b = 6.4, f = .32$; SYK: $w = .25, \beta J = 20, L = 1000$

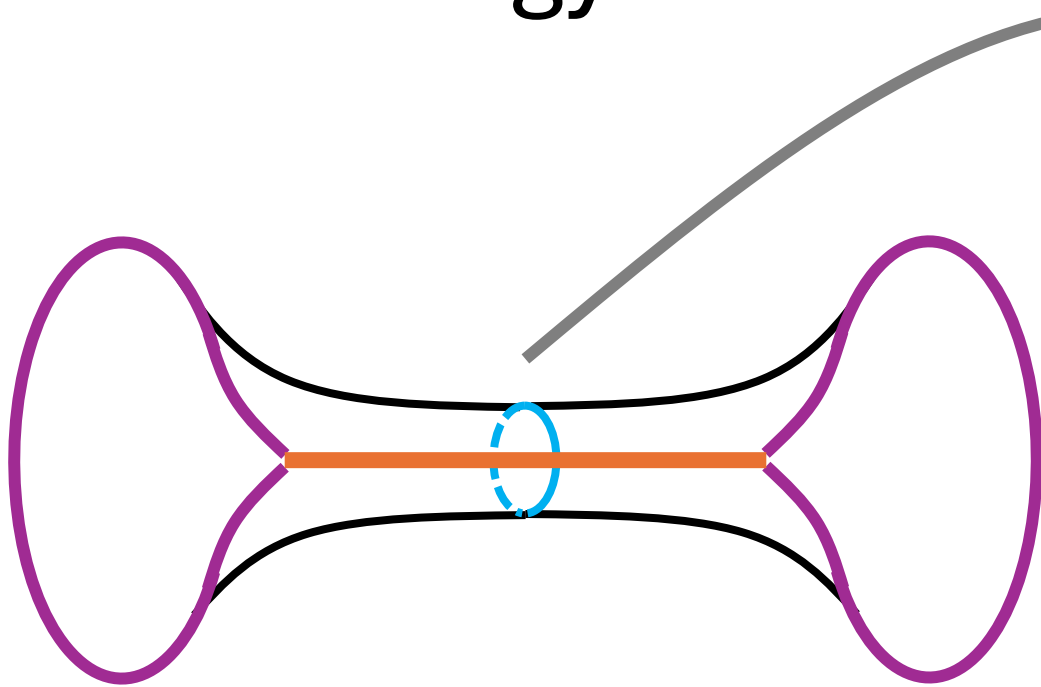
[[Bettaque-S](#), larger beta WIP [Pollack-Milgate-S](#)]

Action data

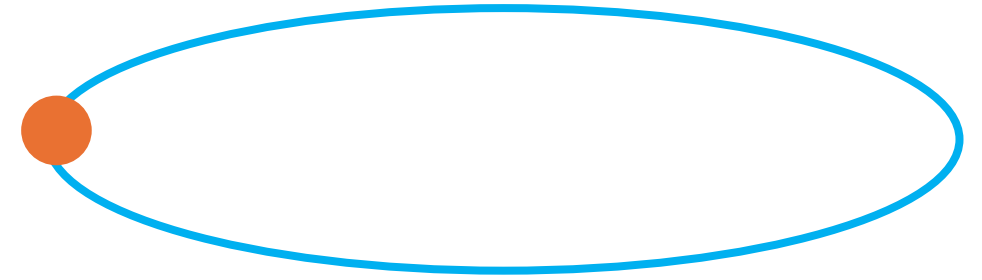
$$I_{\text{JT}} = -S_0\chi - \frac{1}{2} \int d^2x \sqrt{g} \Phi (R + 2) + (\text{boundary terms}) \textcolor{red}{- m\ell_H} + I_{\text{matter}}$$



Cosmology???



Euclidean wormhole with
two asymptotic regions



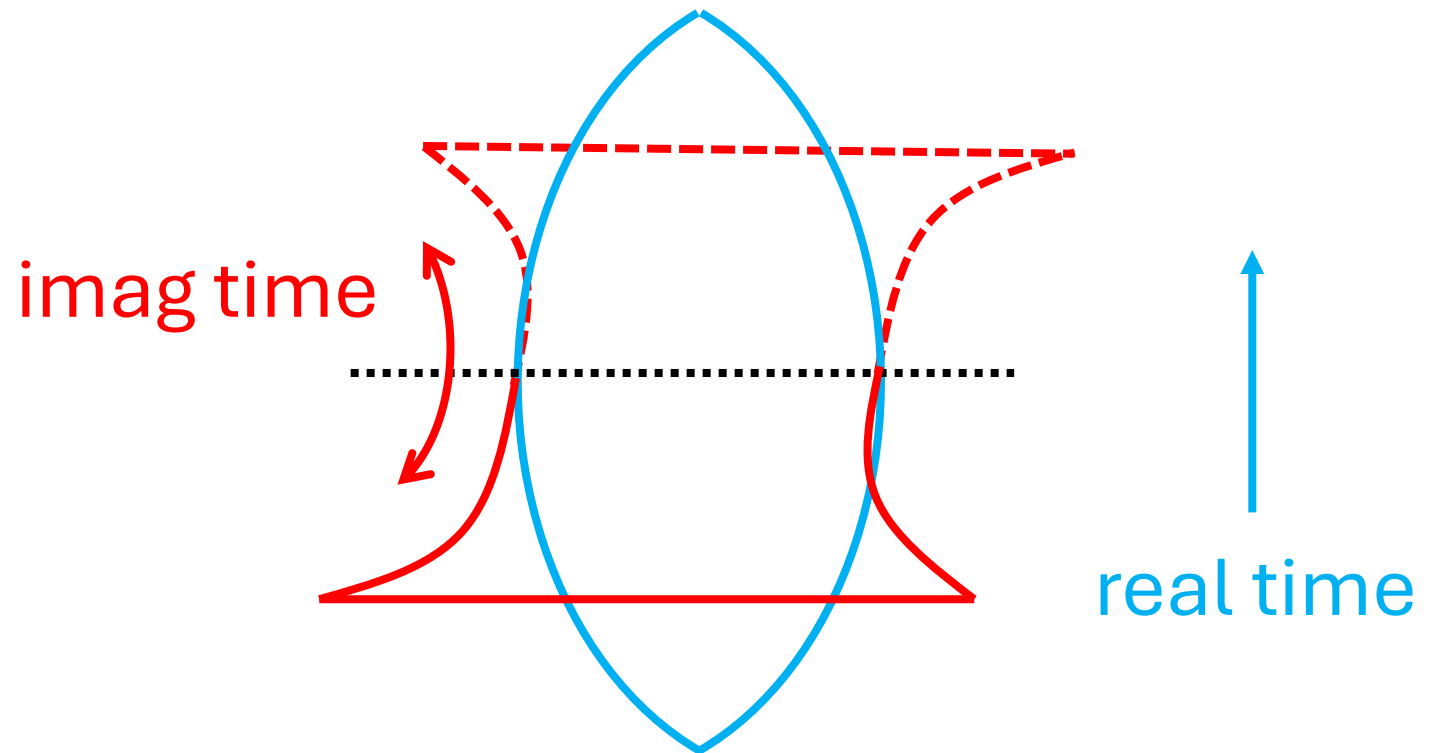
A closed universe consisting of a
mostly empty circle with a heavy
clump of matter

Euclidean wormhole connecting two distinct AdS boundaries
“Wick rotates” into a **Big Bang – Big Crunch cosmology**

Higher dimensional example
(Flat with Neg CC and Rad):

$$a(\rho)^2 = \cosh \rho/t_*$$

$$\rho \rightarrow -it: a(t)^2 = \cos t/t_*$$



[Maldacena-Maoz, McInnes, Antonini-Waddle-Simidzija-Van Raamsdonk-S, ...]

Towards realism?

Flat + radiation + $\Lambda < 0$

$$ds^2 = -dt^2 + a(t)^2(dx^2 + dy^2 + dz^2)$$

$$a(t)^2 = \cos t/t_*$$

symmetric



Big Crunch



Add enough bells and whistles (e.g. scalar fields) → you are here??

Big Bang

time

[more realistic models: Antonini-Waddell-Simidzija-Van Raamsdonk-S, ...]

Outlook

“Standard” outlook

Our questions, again:

- How can we build empty space and wormholes?
- How can we build black holes?
- How can we probe black hole dynamics?
- How can we build cosmologies?

Enormous progress on these questions since I started as a junior faculty in 2017; what will the future look like in ~ 10 more years???

A dream:

Use a quantum computer to simulate and study toy models of quantum cosmology and the Big Bang in a holographic model of quantum gravity

AI!?



Single-Instance Observables in the Sachdev-Ye-Kitaev Model

Brian Swingle

We present a diagrammatic expansion for thermal expectation values of low-weight Majorana strings in a fixed realization of the N -fermion Sachdev--Ye--Kitaev (SYK) model. At each order in the expansion, the observable is expressed as a finite sum of polynomials in the couplings multiplied by temperature-dependent kernels built from the large- N melonic propagator. At fixed βJ , the expansion is organized in powers of $N^{-1/2}$, and each order can be efficiently evaluated with classical resources polynomial in N . The expansion also extends to correlations at arbitrary imaginary times and sufficiently short real times. For the four-body SYK model, we show that the diagrammatic predictions give an excellent account of exact diagonalization results at sizes up to $N = 24$. We also apply the method to the Maldacena--Qi (MQ) model formed from two identical copies of a fixed SYK realization. This enables the identification of a nearby SYK-like Hamiltonian whose thermofield double state has higher fidelity with the Maldacena--Qi ground state than the thermofield double of the original SYK Hamiltonian.

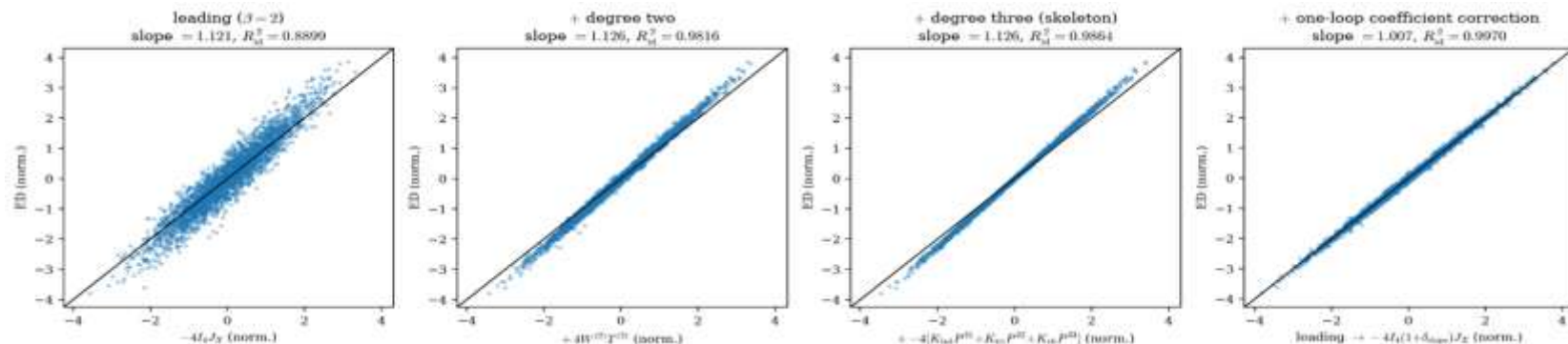
Comments: 78 pages, produced in part with AI tools as described in acknowledgments, 5 minute video abstract [this https URL](https://www.youtube.com/watch?v=...)

Subjects: **High Energy Physics - Theory (hep-th)**; Strongly Correlated Electrons (cond-mat.str-el); Quantum Physics (quant-ph)

Cite as: [arXiv:2609.16130 \[hep-th\]](https://arxiv.org/abs/2609.16130)

(or [arXiv:2609.16130v1 \[hep-th\]](https://arxiv.org/abs/2609.16130v1) for this version)

<https://doi.org/10.48550/arXiv.2609.16130> 



$$\langle \psi_i \psi_j \psi_k \psi_l \rangle = ?$$